# CollTrace: THE DEVELOPMENT OF A WEB-BASED COLLEGE TRACER MANAGEMENT SYSTEM FOR MONITORING OF SEM-ENDER FACULTY REQUIREMENTS

Undergraduate Capstone Project Outline
Submitted to the faculty of the
College of Engineering and Information Technology
Cavite State University
Indang, Cavite

In partial fulfillment
of the requirements for the degree
Bachelor of Science in Information Technology

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# CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements

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An undergraduate capstone project outline submitted to the faculty of the Department of Information Technology, College of Engineering and Information Technology, Cavite State University, Indang, Cavite in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology with Contribution No. _____. Prepared under the supervision of Joy M. Peji

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## INTRODUCTION

Records are essential to academic institutions, serving as reliable evidence of actions, transactions, and institutional decisions (Sebucao-Orias, 2023). Given this role, a tracer management system is of paramount importance in this context, as it transforms the passive storage of files into an active, high-precision monitoring tool. By providing real-time tracking, such a system ensures absolute accountability and administrative transparency. Generally, management of records allows for the immediate identification of pending tasks through automated indicators, ensuring that quality assurance standards are not just met, but are consistently verifiable. For an institution, the importance of this software lies in its ability to safeguard data integrity and streamline the preparation of evidence for institutional audits and global accreditation evaluations.

As a premier institution committed to "Truth, Excellence, and Service," CvSU requires its faculty across various colleges to submit End-of-Semester Requirements to maintain high-quality instruction. On of which is the College of Engineering and Information Technology (CEIT) with five (5) distinct departments, including the

Department of Agricultural and Food Engineering (DAFE), Department of Civil Engineering and Allied Studies (DCEA), Department of Computer, Electronic, and Electrical Engineering (DCEEE), Department of Industrial Engineering and Technology (DIET), and Department of Information Technology (DIT). The College Office serves as the central hub responsible for collecting and archiving these critical academic documents from all faculty members.

Despite the move toward digital solutions, there exists an unsatisfactory condition in the current workflow of CEIT. The college currently manages requirements through a manual process using Google Drive as a repository. While this moved the college away from a purely paper-based system—which was prone to physical loss and damage—Google Drive has proven increasingly insufficient for the growing volume of data. It lacks structured accountability, automated notifications, and real-time tracking capabilities. In the current setup, tracing a specific faculty member's compliance requires time-consuming manual searches through layers of folders, often leading to "digital clutter" where files can be accidentally deleted, moved, or unverified, causing significant delays during urgent audits.

To address these inefficiencies, this researcher proposes the development of CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements. This proposed solution is a dedicated platform designed to replace the generic Google Drive workflow with an automated, centralized, and secure record management environment. The name CollTrace came from "College Trace" which is the main purpose of the proposed system. Unlike standard cloud storage, CollTrace features a custom dashboard with automated status indicators using colored labels for early, on-time, and late submissions, and real-time countdowns to deadlines. By digitizing the archival process and offering role-specific access for administrators, department chairpersons, and faculty staff, the system ensures that CEIT can

generate accurate compliance reports in seconds, directly supporting the university's mission of achieving global excellence in academics and administrative precision.

**Project Context**

To mitigate the physical risks, CEIT transitioned to using Google Drive as a digital repository, organizing faculty submissions through a structured folder system. While this shift addressed the immediate concerns of physical document loss, it introduced a new set of limitations inherent to general-purpose cloud storage platforms. Specifically, Google Drive lacks a built-in mechanism to automatically track submission compliance, notify administrators of missing requirements, or restrict user actions that could compromise file integrity. Files can be accidentally deleted, moved, or misnamed by users with edit access, resulting in digital disorganization that makes retrieval increasingly difficult as the volume of submissions grows (Urmeneta, 2025). Furthermore, tracing a specific faculty member's compliance status often requires a time-consuming manual search through multiple nested folders, which is incompatible with the fast-paced demands of modern academic administration.

As shown in Appendix Figure 1, poor documentation storage in CEIT can be attributed to factors across four categories. Due to their heavy workload, faculty members sometimes submit files without following a standard naming convention, and no required folder structure or submission deadline enforcement mechanism is in place. The Dean's office manually checks each folder, faculty members are often unaware of missing requirements, and no dedicated staff is assigned for compliance monitoring. Google Drive has no access control per faculty, files can be deleted or moved by any user, and no automated backup exists for submitted documents, with

storage also being limited. The high volume of records, the absence of a unified checklist per faculty role, and inconsistent file formats across the five departments all contribute to the problem of poor documentation storage.

As presented in Appendix Figure 2, the lack of real-time tracking in the college's submission process is attributed to four categories of contributing factors. Compliance status is only checked at the sem-ender period, no dashboard exists to show per-department compliance, and status updates rely on manual folder inspection. Chairpersons manually check faculty folders, no person is assigned to monitor submission status, and faculty are not reminded of their pending requirements. Google Drive has no built-in compliance tracker, no automated notification system exists for missing submissions, and shared Drive links tend to become broken over time. Lastly, the high volume of faculty across five departments, the time-critical nature of sem-ender deadlines, and the differing requirement sets per department all make real-time tracking increasingly necessary.

Appendix Figure 3 presents the factors contributing to insufficient manual verification of faculty submissions in CEIT. The Dean's office opens each folder individually to verify submissions, no proof-of-submission record is kept per faculty, and the verification process is repeated every semester without any historical log for reference. Staff spend hours verifying submissions per department, errors in manual checking can lead to missed submissions, and personnel may rely on memory to assess compliance. No system exists to auto-generate compliance reports, Google Drive cannot flag incomplete submissions, and no audit trail is maintained for submitted documents. Additionally, the requirements list may differ each semester, no

digital checklist is available to track submissions per faculty, and submitted files are not locked after compliance has been confirmed.

**Objectives of the Study**

This study aims to allow the College of Engineering and Information Technology to organize their documentation using a Web-Based Record Management and Tracking System. In addition, the system helps to have a secure and transparent document storage or database.

Specifically, the study aims:

1. To design a module that

   1.1 allow the Faculty Staff upload documents for specific folders created by the Administrators;

   1.2 support real-time data access for the Department Head / Chairperson; and'

   1.3 generate an accurate summary or report analysis of submissions using percentages for Administrators and Department chairpersons.

2. To evaluate and test the system based on ISO 25010 software quality standards in terms of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability;
3. To implement a system for the College of Engineering and Information Technology.

**Time and Place of the Study**

This study was conducted at Cavite State University - Main Campus located in Indang, Cavite. The development and documentation phase of the study began in

April 2026 and is expected to be completed by December 2026. This duration encompasses the system analysis, design, coding, testing, and final deployment phase.

**Scope and Limitation of the Study**

The purpose of the Web-Based Tracking and Record Management System is to provide the College of Engineering and Information Technology (CEIT) with a working web-based requirements tracking and record management system. This will be accomplished by developing a centralized database where all end-of-semester faculty requirements such as grades, Tables of Specifications (TOS), and other academic documents submitted by faculty from various departments can be stored, organized, and monitored collectively.

The study focuses on addressing the inefficiencies and lack of structured accountability currently experienced when using Google Drive for document storage and retrieval.

The target beneficiaries of the study include Dean’s office staff, department chairpersons, and faculty staff of five departments of CEIT: DAFE, DCEA, DCEEE, DIET, and DIT.

The system is described as a platform developed to provide faculty requirement management support to administrators, department chairpersons, and faculty members during the course of a semester.

The web-based management and tracking system has the following capabilities:

1. The administrative user can create user accounts. By the power of a high user level, the administrator can create accounts using the university email and basic information.
2. Automatically closed End-of-Semester requirement folder. At the end of the semester, there will be an ability to close out and archive the previous semester's information into a format that is read-only to access for future semesters, and reset the system for the next semester.
3. The administrative user can create folders. For every End-of-Semester the administrator has an ability to create a folder named for a specific academic year and semester.
4. Visible label color indicators for faculty staff requirements submission status. The department chairpersons can check the status of all requirements passed by the faculty staff. Yellow for not yet submitted, Green for early-submitted or submitted, and Red for late.
5. Department chairpersons can assign a faculty member to upload a specific requirement.
6. Faculty members will have access to see the requirements assigned to them.
7. Faculty members were notified that they didn’t submit the specific requirement yet.

The importance of this system lies in its rationale, timeliness, and relevance to the university's goal of excellence. By transitioning from a generic storage platform to a custom web-based system, CEIT ensures that its institutional records are secure, easily retrievable, and audit-ready. This highlights the significance of the project as it directly solves existing problems related to manual tracking and the disorganized nature of shared cloud storage. Ultimately, the end result will be improved consistency, accountability, and organization with regard to CEIT's

management of academic records, benefitting the college as a whole, the department chairpersons, and the faculty staff.

The system will feature distinct levels of user access. Faculty members will have a custom dashboard to view pending requirements, track countdowns to deadlines, and submit documents such as student outputs, lab manuals, and exams. Administrators and department chairpersons will have access to an automated dashboard for monitoring compliance percentages, viewing real-time submission statuses, and managing user accounts. The proposed web-based system will also have 3 modules: Account Management, Record/Data Management, Report/Summary Management. The system is a web-based system and viewable in mobile through a browser.

Account Management handles all user-related administrative tasks to ensure secure system access. This module provides the College Office Administrator with comprehensive control, including the ability to create new user accounts, assign specific user roles, and view all existing accounts within the system. Additionally, the information of the user cannot be changed by the owner of the account.

Record / Data Management serves as the core operational area for handling academic requirements. This module is structured hierarchically to support the different levels of college authority. The Department Chairperson is granted the capability to view uploaded documents for their respective departments while the Administration is granted the capability to view uploaded documents for all departments to ensure compliance. Meanwhile, Faculty Staff are provided with the primary function to upload documents, transitioning the college from manual, paper-based submissions to a more efficient digital repository.

Report / Summary Management provides high-level reporting capabilities for administrative oversight. This module enables the College Office Administrator to

view all uploaded documents collectively, check the real-time status of specific submissions, and track the overall submission percentage. This ensures that the web-based system can generate accurate compliance summaries for certification purposes in a matter of seconds.

The study is subject to several limitations beyond the control of the researchers. The system's functionality is dependent on the stability of the user's internet connection and server uptime, as it is a web-based architecture. The data to be managed is limited only to academic End-of-Semester requirements. However, the other administrative or personal faculty records are outside the scope of this tracking system. Furthermore, the system is specifically designed for the internal workflow of CvSU-CEIT and may not be immediately applicable to other colleges with different departmental structures or submission protocols.

**Conceptual Framework of the Study**

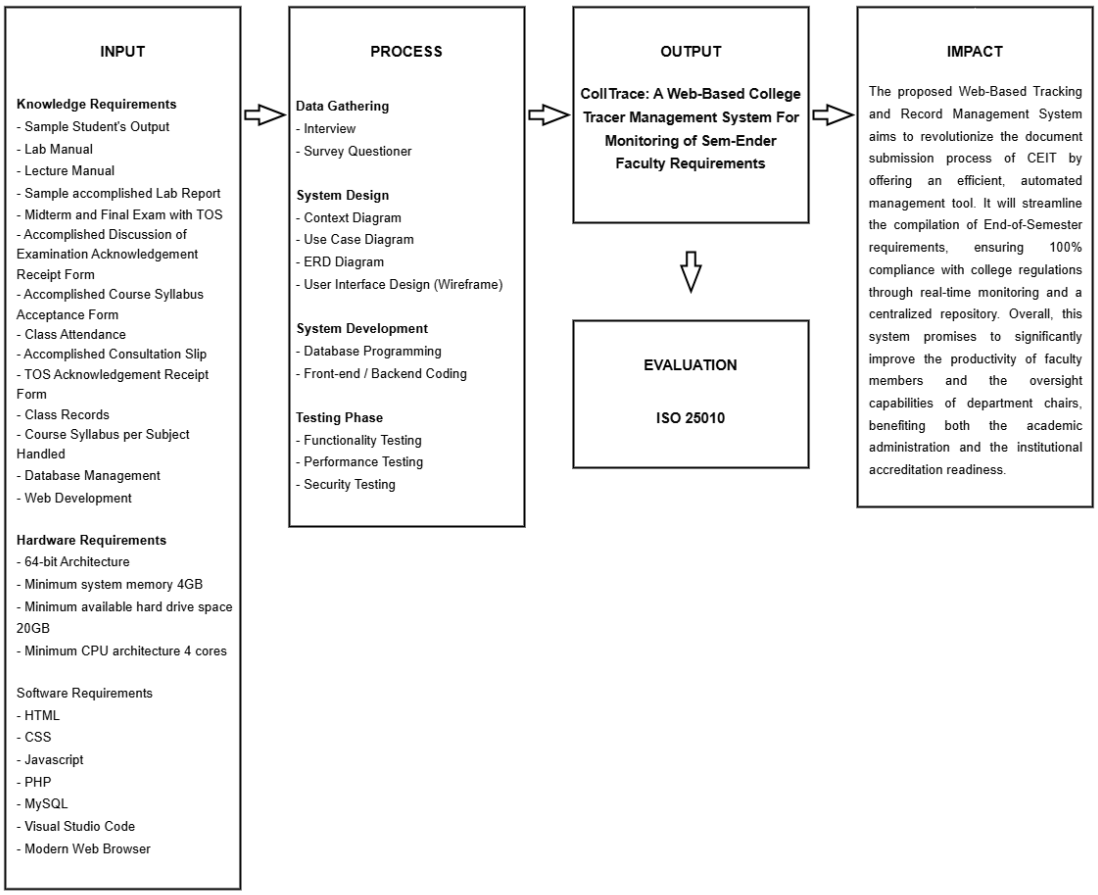



Figure 1. Conceptual Framework of a Web-based Record Management and Tracking System for Cavite State University - CEIT

The conceptual framework follows an Input-Process-Output-Impact (IPOI) model, integrated with an evaluation stage to ensure system quality. The Input Phase encompasses the foundational requirements necessary for development, categorized into knowledge, hardware, and software. Knowledge requirements include a comprehensive list of academic documents—such as student outputs, lab manuals, syllabi, and class records—alongside technical proficiency in web development and database management. These are supported by specific hardware specifications, including a 64-bit architecture with at least 4GB of RAM, and a software stack

featuring HTML, CSS, JavaScript, PHP, and MySQL, utilized within the Visual Studio Code environment.

The Process Phase details the systematic progression of the project, beginning with data gathering through interviews and surveys to identify user needs. This leads into the system design stage, where the architecture is mapped via context, use case, and ERD diagrams, alongside UI wireframes. During system development, the researchers engage in database programming and full-stack coding. This is followed by a rigorous testing phase covering functionality, performance, and security to ensure the system is robust. Crucially, the process includes an evaluation component where the system is measured against the ISO 25010 standard to validate its technical quality and usability.

The Output Phase marks the realization of the project: "CollTrace: A Web-Based Record Management and Tracking System." This final product serves as the centralized repository designed to provide a transparent and organized method for managing university records. Finally, the Impact Phase outlines the long-term benefits of the system. By automating the document submission process for CEIT departments, CollTrace aims to ensure 100% compliance with college regulations through real-time monitoring. The ultimate goal is to enhance faculty productivity, empower department chairs with better oversight, and strengthen the institution’s overall readiness for academic accreditation.

**Definition of Terms**

- **Account Management** - a specialized module that handles all user-related administrative tasks to ensure secure and authorized access to the system.

- **Accreditation** - framed around how CollTrace prepares CEIT for institutional audits and faculty clearance.

- **Activity Log** – the system feature that records all user actions for transparency and accountability.
- **Administrator** - refers to the college office staff with high-level access who manages user accounts, creates academic year folders, and oversees the monitoring of all departmental submissions.
- **Agile Development Methodology** – the iterative software development framework utilized in this study, characterized by continuous testing, defect fixing, and requirement refinement across multiple development phases to ensure the system meets evolving CEIT needs.
- **Centralized Repository** - a unified digital database where all end-of-semester faculty requirements are stored and organized, serving as the single source of truth for the College of Engineering and Information Technology.
- **CollTrace** - name of the system that came from the phrase College Trace.
- **Compliance Status** - the real-time categorization of a faculty member's submission, visually represented by colored labels: yellow for pending, green for on-time or early, and red for late submissions.
- **Dashboard** – the personalized landing page each user role sees after logging in.
- **Database** – the MySQL-based backend that stores all submission records, user credentials, and folder data.
- **Deadline** – the submission cutoff enforced by the system through countdown timers and label color shifts.
- **Department Chairperson** - The academic official responsible for assigning faculty requirements, monitoring the real-time submission status of their specific department, and viewing compliance percentages.
- **Digital Clutter** - The disorganized state of file storage common in shared cloud environments, characterized by accidentally deleted, moved, or misnamed files that make retrieval difficult.

- **Faculty Staff** - The user group responsible for uploading required academic documents to specific folders and accessing a custom dashboard to track their own pending requirements and deadlines.
- **Folder** – the structured digital directory created per academic year that auto-generates across all 5 departments.
- **Google Drive -** The current digital repository used by CEIT that serves as a baseline for the study, representing a manual, non-automated file storage system that lacks structured tracking and notification features.
- **ISO 25010 Standard** – The international software quality framework used to evaluate the system based on eight specific characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.
- **Record Management -** The digital process of storing, organizing, and archiving institutional documents in a centralized repository, replacing the current manual reliance on Google Drive and paper-based folders.
- **Sem-Ender Requirements -** The specific set of academic documents that faculty members must submit, which includes Student Outputs, Lab Manuals, Exams, Tables of Specifications (TOS), and grade sheets.
- **Tracking System** - A specialized software functionality that provides real-time monitoring of document submissions, including automated countdowns to deadlines and percentage-based compliance reporting.
- **User Account** – the registered profile created by the Admin that grants role-specific access to the system.
- **Web-based Architecture -** The system's structural design that utilizes a client-server model, allowing users to access the tracking dashboard and upload documents through a standard web browser on both desktop and mobile devices.

# REVIEW OF RELATED LITERATURE

The design and development of CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements is grounded in a broad body of existing literature and studies. This chapter synthesizes relevant foreign and local works that collectively shed light on the importance of web-based record management systems, digital compliance tracking, automated notification mechanisms, and information systems in academic institutions. The literature is organized into four sections: Foreign Related Literature, Local Related Literature, Foreign Related Studies, and Local Related Studies.

## Foreign Related Literature

The widespread digitization of academic administration has been a prominent area of inquiry in the international scene. A growing body of foreign literature highlights the limitations of manual and generic record systems and the clear advantages of purpose-built web-based platforms that automate tracking, ensure compliance, and centralize institutional data.

### Web-Based Centralized Tracking and Record Management

Karpe et al. (2025) conducted a comprehensive review of existing web-based centralized tracking systems and found that these platforms significantly enhance transparency, accuracy, and efficiency in managing academic records. The authors noted that such systems enable real-time monitoring, reduce manual errors, and empower administrators and educators to adopt data-driven strategies. Crucially, their review pointed out that systems relying on unstructured storage lack the automated oversight needed for large-scale academic compliance tracking — a

limitation directly addressed by the proposed CollTrace system. The study also recommended integrating cloud-based solutions and strengthening data security protocols, both of which are incorporated in the technical direction of the proposed system.

Similarly, Anichebe and Asogwa (2025) explored the development of a web service framework for the secure and efficient transmission of academic records across institutions. Their work demonstrated that a properly structured web framework can standardize how academic data is handled, reducing redundancy and improving retrieval speed. While the focus of their study was on inter-institutional transmission rather than internal departmental tracking, the underlying principle — that structured, role-based access to academic documents improves institutional efficiency — is directly applicable to the CollTrace System. Faculty staff, department chairpersons, and administrators in the proposed system each have defined roles that mirror this framework of controlled, structured access.

Amanze et al. (2022) designed and implemented a cloud-based document management system for academic institutions, demonstrating that cloud architecture enables scalable, centralized storage that reduces reliance on physical files and local servers. The study highlighted the importance of defining user roles and access privileges as a means of safeguarding institutional documents. For CollTrace, this study reinforces the design decision to implement distinct access levels for administrators, department chairpersons, and faculty staff, ensuring that each user type can only perform actions appropriate to their role within the college.

Nurpandi and Koswara (2023) analyzed and designed an Electronic Document Management System (EDMS) for the Faculty of Engineering at Suryakancana University. Their study established that an EDMS significantly reduces document processing time and eliminates the risks associated with physical record-keeping, such as loss, damage, and unauthorized access. The findings align

closely with the problems identified in CEIT's current use of Google Drive, particularly the risks of accidental file deletion and disorganized folder structures that the proposed CollTrace system aims to resolve through a more controlled digital environment.

**Accreditation, Quality Management, and Compliance Documentation**

Iqbal et al. (2025) examined the role of accreditation and quality management initiatives in driving institutional performance in higher education. Their mediation analysis revealed that structured record-keeping and evidence-based compliance documentation are central to meeting accreditation standards. This is particularly relevant to CollTrace, which aims to make end-of-semester faculty requirements readily available for audit and accreditation purposes. The ability to generate compliance reports in a matter of seconds, as described in the system's features, directly supports the kind of documentation readiness that Iqbal and colleagues identify as critical to quality-driven academic institutions.

**Management Information Systems in Higher Education**

Nurhidayah and Muliansyah (2023) reviewed management information systems in private universities and established that well-implemented digital systems significantly improve institutional productivity and administrative decision-making. The authors found that MIS reduces workload and centralizes data storage, though they also cautioned that the absence of trained personnel and high implementation costs remain persistent challenges. For CollTrace, this finding underscores the importance of designing an intuitive interface that minimizes the learning curve for faculty staff, chairpersons, and administrators who will interact with the system.

### Automated Notification and Deadline Compliance

Motz et al. (2021) conducted a randomized controlled trial on automated educational nudges designed to reduce missed assignments among college students. Their results confirmed that automated, timely alerts — particularly those delivered through a familiar digital interface — significantly lowered the rate of missed deadlines. While the study focused on students rather than faculty, the behavioral mechanism it documents applies equally to faculty submission compliance in an institutional setting. The color-coded status labels used in CollTrace (yellow for pending, green for submitted, and red for late) function as a visual form of the same kind of nudge, giving faculty members and administrators an immediate at-a-glance cue about submission urgency.

## Local Related Literature

In the Philippine context, academic literature has increasingly recognized the inadequacies of manual and fragmented document management practices in educational institutions. Local scholars have documented both the extent of these gaps and the demonstrated benefits of transitioning to structured digital systems.

### Records Management Practices in Philippine Academic Institutions

Sebucao-Orias (2023) assessed the extent of records management practices in a state university in the Philippines using a descriptive research approach guided

by ISO 15489 standards and Republic Act 9470. The study revealed that while records monitoring was generally upheld, records control was only moderately maintained, and the university's weakest practices included disaster recovery planning, archival preservation, and digital storage and backup systems. The absence of structured digital control highlighted in this study is mirrored in the CEIT office's current experience with Google Drive, where files can be accidentally deleted, moved, or misnamed without accountability.

**Education Management Information Systems and Accreditation**

Villar (2020) conducted a mini-review of Education Management Information Systems (EMIS) in the Philippine educational context and found that centralized data systems helped institutions meet accreditation standards set by AACCUP and PAASCU more efficiently. However, a recurring finding was that many local academic systems remain isolated, operating independently across departments without full integration. Villar recommended the development of a unified database that connects all administrative units within an institution — precisely the direction that CollTrace pursues by providing a single, role-based platform accessible to administrators, chairpersons, and faculty staff across all five CEIT departments.

**Electronic Document Management Systems in Philippine Higher Education Institutions**

Pagayonan (2022) developed an E-Document Management System for Northern Iloilo State University to address ISO 9001:2015 Quality Management System compliance issues. The study found discrepancies in the tracking, preservation, and forwarding of memoranda and files, as well as problems with document loss and redundancy under the institution's existing manual system. The

resulting digital system was evaluated by 165 participants using a developmental-descriptive research design, and results confirmed significant improvements in usability and performance. This study is particularly relevant to CollTrace because it presents a directly analogous scenario: a Philippine state university that identified critical gaps in its document control mechanisms and addressed them through a purpose-built digital management system.

Angala et al. (2023) developed and implemented a web-based Document Management System (DMS) at Ilocos Sur Polytechnic State College, Tagudin Campus, demonstrating that a centralized digital environment for capturing, storing, organizing, and retrieving documents significantly addressed the institution's challenges with physical document accumulation and inefficient retrieval. The study found that system design and user engagement were the primary drivers of successful DMS adoption, rather than individual demographic factors among users. For CollTrace, this finding reinforces the importance of an intuitive, role-differentiated interface that facilitates faculty and administrator engagement without extensive technical training.

**Digital Faculty Monitoring and Centralized Data Management**

Mondoza (2022) proposed a framework for digitalizing faculty monitoring and evaluation through an online system designed to centralize faculty performance data and improve evaluation speed and accuracy. The study established that centralized data management significantly improves how administrators track and evaluate faculty performance. While Mondoza's framework focused primarily on performance evaluation rather than document submission tracking, the principle of consolidating faculty-related data into one accessible platform is directly foundational to CollTrace.

The proposed system extends this principle by applying it specifically to end-of-semester requirements, with role-based access designed to match the administrative hierarchy of a Philippine college.

## Foreign Related Studies

Beyond theoretical literature, several foreign studies have documented the practical implementation and outcomes of web-based systems in academic record management, document tracking, and compliance monitoring. These studies provide empirical grounding for the design decisions embedded in CollTrace.

### Web-Based School Record Management and Performance Outcomes

Ugalingan and Antonio (2025) developed and evaluated SRSys, a web-based school record management system, through a quasi-experimental comparison between the manual method and the automated platform. Their findings were striking: error rates dropped from an average of 7.4 per task to just 1.6, while task completion time fell from 228.9 seconds to 69.4 seconds. User satisfaction ratings consistently exceeded 4.85 on a five-point scale, and paired t-tests confirmed the statistical significance of improvements in both accuracy and speed. For CollTrace, this study is particularly foundational because it provides direct, measurable evidence that transitioning from manual document handling to a web-based system yields concrete, quantifiable benefits — the same transition CEIT is seeking to make by moving from Google Drive to the proposed system.

**Integrated Systems for Attendance and Assessment Submission**

Othman et al. (2022) developed an integrated web-based system for recording student attendance and managing assessment submissions at Universiti Teknologi MARA in Malaysia. The system was built using the System Development Life Cycle with a prototyping approach and tested with 97 students over one semester. Results showed high acceptance ratings with mean scores between 3.89 and 3.95 on a four-point scale, and a significant reduction in the clerical workload borne by lecturers. This study directly parallels the submission management module of CollTrace, which similarly aims to reduce the administrative burden on CEIT department staff by automating the receipt, categorization, and monitoring of required academic documents.

**Compliance Tracking for Teaching-Learning Deliverables**

Salisi and Balahadia (2025) developed EduComply, a progress tracking system specifically designed for monitoring the submission of teaching-learning deliverables in an academic institution. The system used a Decision Support System (DSS) framework and demonstrated improved compliance and more organized record management. EduComply is perhaps the most thematically aligned foreign study to CollTrace, as both systems target the specific challenge of tracking whether faculty members have submitted required academic documents. The deliverables tracked in EduComply — syllabi, self-paced learning modules, tables of specifications, and exams — closely mirror the end-of-semester requirements that CollTrace is designed to manage within CEIT.

### Dashboard-Centric Systems for Real-Time Submission Monitoring

Mawarni et al. (2022) developed a management information system specifically for monitoring academic report submissions, featuring a dashboard that allows unit heads to see real-time progress indicators for each department's compliance. Using the Waterfall Model and black-box testing, the study demonstrated that administrators could view at a glance which units had not yet submitted reports. This dashboard-centric design is directly reflected in CollTrace's reporting module, where administrators and department chairpersons can monitor submission percentages and identify non-compliant faculty members in real time.

## Local Related Studies

A number of local studies conducted in Philippine institutions offer direct empirical precedent for the system being developed. These studies demonstrate that web-based document tracking and management systems are both feasible and effective in the Philippine academic context, and they provide concrete design models applicable to the development of CollTrace.

### Document Tracking and Administrative Transparency

Piloton et al. (2023) developed a records tracking management system for Surigao del Norte State University that integrated QR code technology to monitor the movement of documents across offices. Using the Software Development Process methodology, the system improved administrative transparency by logging every instance of document movement, making the flow of paperwork visible to administrators at all times. The study found that accountability increased substantially when each document's location and status was automatically recorded rather than

tracked manually. For CollTrace, this principle of automated, timestamped status recording is replicated through its real-time submission status labels, providing a similarly transparent audit trail for every requirement submitted by faculty members.

**Digital Document Management in Philippine State Universities and Colleges**

Abisado et al. (2020) developed and validated a document tracking model for Philippine state universities and colleges, demonstrating that a purpose-built system could facilitate the monitoring and tracking of in-process documents more effectively than manual methods. The study used the Software Development Process as its development basis and validated the system with 40 office personnel, confirming its acceptability in terms of user interface and functionality. This study is particularly foundational for CollTrace because it addresses the same institutional setting — Philippine state universities — and the same core problem: the lack of a structured, automated mechanism for tracking document submissions across administrative units.

**Progress Monitoring and Compliance in Philippine Academic Institutions**

Salisi and Balahadia (2025), in their development of EduComply at Laguna State Polytechnic University, provided a concrete local model for faculty deliverable tracking within a Philippine higher education institution. Their study's Agile-based development approach and emphasis on real-time compliance dashboards offer a directly applicable precedent for the development process and feature set planned for CollTrace. The local institutional context of EduComply — including the challenge of monitoring multiple types of faculty submissions across a college setting — closely mirrors the operational environment of CEIT, where five departments each require their faculty to submit distinct sets of end-of-semester requirements.

Table 1. Related studies comparison table

| **MODULE / FEATURES** | **S1** | **S2** | **S3** | **S4** | **S5** | **S6** | **S7** | **S8** | **PROPOSED STUDY** |
|---|---|---|---|---|---|---|---|---|---|
| Centralized Document Storage | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| Role-Based Access Control | ✓ | | ✓ | ✓ | | | ✓ | ✓ | ✓ |
| Submission Status Tracking | | | ✓ | | ✓ | | | | ✓ |
| Automated Notifications / Deadlin | | | | | | ✓ | | | **sdc** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| e Reminders | | | | | | | | | |
| Compliance Report Generation / Dashboard | ✓ | | | | ✓ | | ✓ | | ✓ |
| Document Tracking and Accountability | | | | | | | ✓ | ✓ | ✓ |
| **Methodology** | Quasi-Experimental | Prototyping / SDLC | DSS Framework | Descriptive | Waterfall Model | Experimental | Agile / RAD | Developmental-Descriptive | Agile |

**Note:**

S1: Ugalingan and Antonio (2025)

S2: Othman et al. (2022)

S3: Salisi and Balahadia (2025)

S4: Pagayonan (2022)

S5: Mawarni et al. (2022)

S6: Haqimi and Kusuma (2023)

S7: Abisado et al. (2020) / Piloton et al. (2023)

S8: Angala et al. (2023)

**Synthesis**

The studies were evaluated according to the key modules that form the foundation of CollTrace: centralized document storage, role-based access control, submission status tracking, automated notifications and deadline reminders, compliance report generation and dashboard monitoring, and document tracking and accountability. Among these features, centralized document storage appeared most consistently across the reviewed studies, with S1, S2, S3, S4, S5, S7, and S8 all incorporating this capability in varying forms. This pattern confirms that consolidating records into a single accessible repository has become a standard expectation for web-based academic systems. Role-based access control, which CollTrace uses to distinguish the responsibilities of faculty members, department chairpersons, and administrators, was present in S1, S3, S4, S7, and S8, reflecting how institutional systems generally require tiered permissions to function properly within a hierarchical organization.

Submission status tracking was identified in S3 and S5, both of which involved monitoring whether specific deliverables had been submitted or completed within a given timeframe. This feature is central to CollTrace since the system is built

precisely to address the difficulty of knowing which faculty members have complied with end-of-semester requirements at any given point. Automated notifications and deadline reminders, on the other hand, were only found in S6, which used the Telegram API to alert lecturers and students of approaching deadlines. While the mechanism differs from what CollTrace employs, the study confirms that timely automated alerts have a measurable positive effect on compliance behavior in academic environments.

Compliance report generation and dashboard monitoring were present in S1 and S5, which featured summary or reporting tools that gave administrators visibility over the state of submissions within their institution. CollTrace incorporates this through its reporting module, where submission percentages can be viewed in real time by those in oversight roles. Document tracking and accountability, which refers specifically to logging the movement and status of submitted documents, was addressed in S7 and S8, both of which demonstrated that automatic recording of document actions significantly improves institutional transparency. This feature is also built into CollTrace to ensure that every submission action is recorded and traceable. In terms of methodology, each study took a different approach, ranging from quasi-experimental and Waterfall-based to DSS frameworks and developmental-descriptive designs. The proposed study adopts the Agile Software Development Life Cycle, which supports iterative development and allows the system to be refined continuously based on feedback gathered throughout the process.

Furthermore, the body of literature and research analyzed in this chapter leads to a singular and clear conclusion: academic institutions that stick to manual

processes or loose digital setups for record-keeping inevitably struggle with accountability, efficiency, and meeting compliance standards. Whether looking at global trends or local Philippine studies, there is a strong consensus that web-based platforms — specifically those featuring centralized storage, hierarchical access levels, and automated monitoring — offer a definitive advantage over older methods.

Empirical data from researchers such as Ugalingan and Antonio (2025), Salisi and Balahadia (2025), and Othman et al. (2022) support this with measurable outcomes, showing that purpose-built systems drastically reduce human error and administrative workload. Within the Philippine setting, the works of Sebucao-Orias (2023) and Villar (2020) highlight a pressing need for change, as many state universities still deal with fragmented systems and weak record controls. More directly, Pagayonan (2022) and Angala et al. (2023) demonstrate that purpose-built electronic document management systems have already been developed and validated in Philippine state colleges, providing concrete design precedents that CollTrace builds upon. Abisado et al. (2020) and Piloton et al. (2023) further confirm that automated document tracking in Philippine state university contexts is both feasible and effective.

Ultimately, these studies do more than just provide a background; they lay out the exact blueprint for why CollTrace is necessary. The proposed system is a direct answer to the flaws identified in current workflows, providing the College of Engineering and Information Technology at Cavite State University with a structured and automated tool designed specifically to handle the high-pressure demands of end-of-semester document tracking.

# METHODOLOGY

In this chapter, the methods of research that were developed will be outlined in detail as used by the researchers. Various types of techniques were used by the researchers to collect and analyze data through a structured and systemized process including steps, system designs, requirements and sub-phases. These techniques were used to manage, control and evaluate the process of research appropriately. This chapter will also describe how the information and data from this research will be used to address each of the identified problems and meet the overall objectives of system development.

## Requirement Analysis

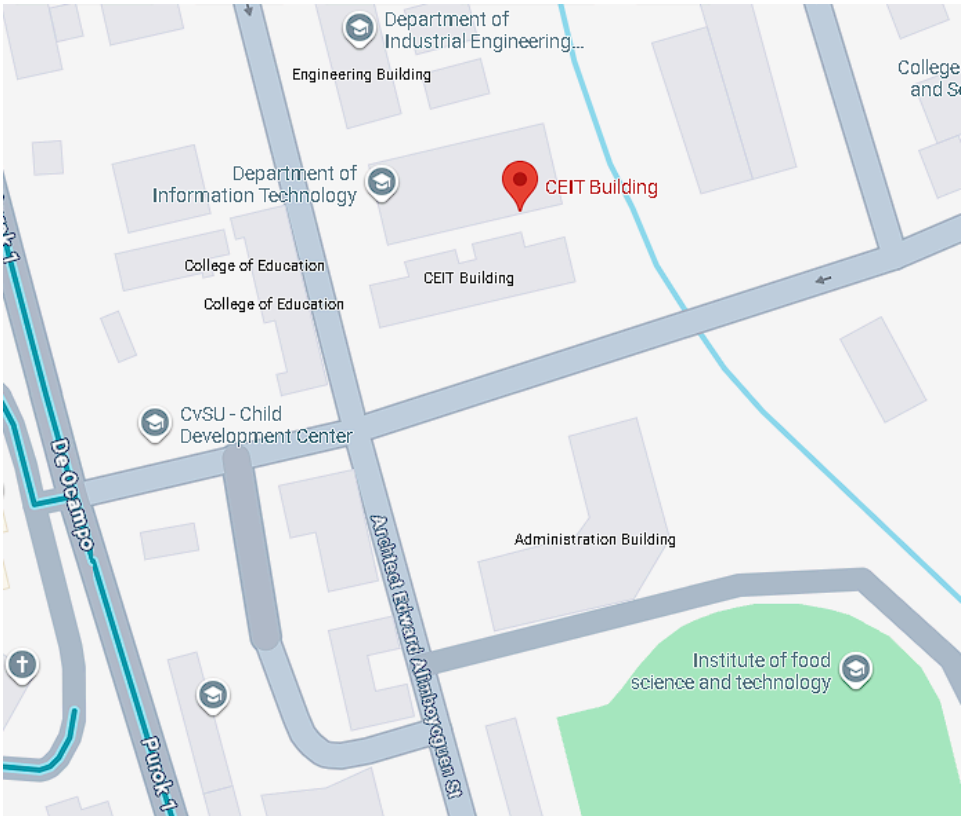



Figure 2. Cavite State University CEIT office building location

Figure 2. Illustrate the map indicating the established location of the CEIT office building at Cavite State University - Don Severino Delas Alas Campus in Indang, Cavite.




Figure 3. Organizational Chart

As illustrated in Figure 3, the researchers conducted a comprehensive analysis of the existing manual yet digital file submission processes within the College of Engineering and Information Technology (CEIT). This stage focused on identifying the specific administrative needs of the following key personnel:

- System Administrators: To establish the framework for academic school years and standardize the required document templates.
- Department Chairpersons: To monitor real-time submission progress and manage faculty compliance across different departments, such as DIT, DIET, DCEE, DAFE, and DCEA.
- Faculty Staff: To streamline the uploading of necessary academic requirements into organized digital folders.

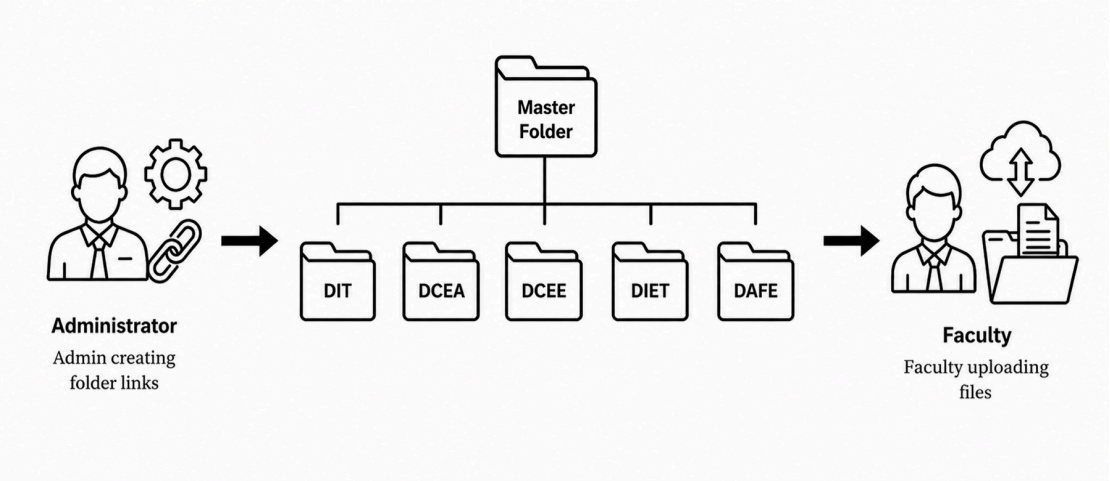



Figure 4. Current technical situation of tracking submission

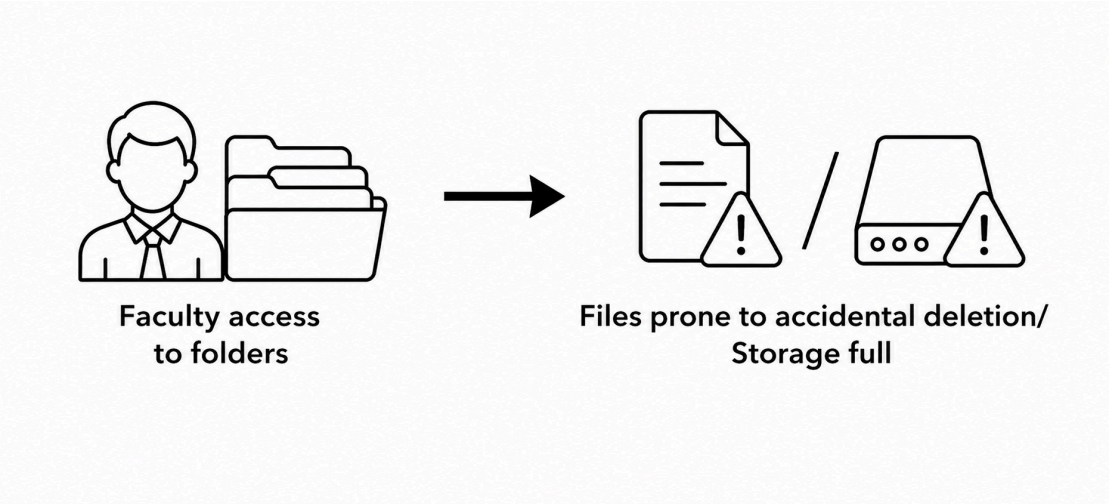



Figure 5. Current technical situation of document management

In Figure 4, the current technical situation of tracking submissions is depicted, where the administrator manually generates Google Drive links and creates specific folders for each department. This is followed by Figure 5, which illustrates the current technical situation of document management, which shows the manual process of faculty members uploading files to shared folders that are prone to accidental deletion and limited storage capacity.

**Requirement Documentation**

The researchers need to identify the requirement documents, specifically the modules of the system, use case diagrams and activity diagrams that show the capabilities of the software to be developed.

**Functional Requirements**

The CollTrace: A Web-Based Record Management and Tracking System consists of three primary modules such as: account management module, record/data management module, and report/summary management module. These modules are designed to work together to automate the submission and tracking of academic requirements within the college.

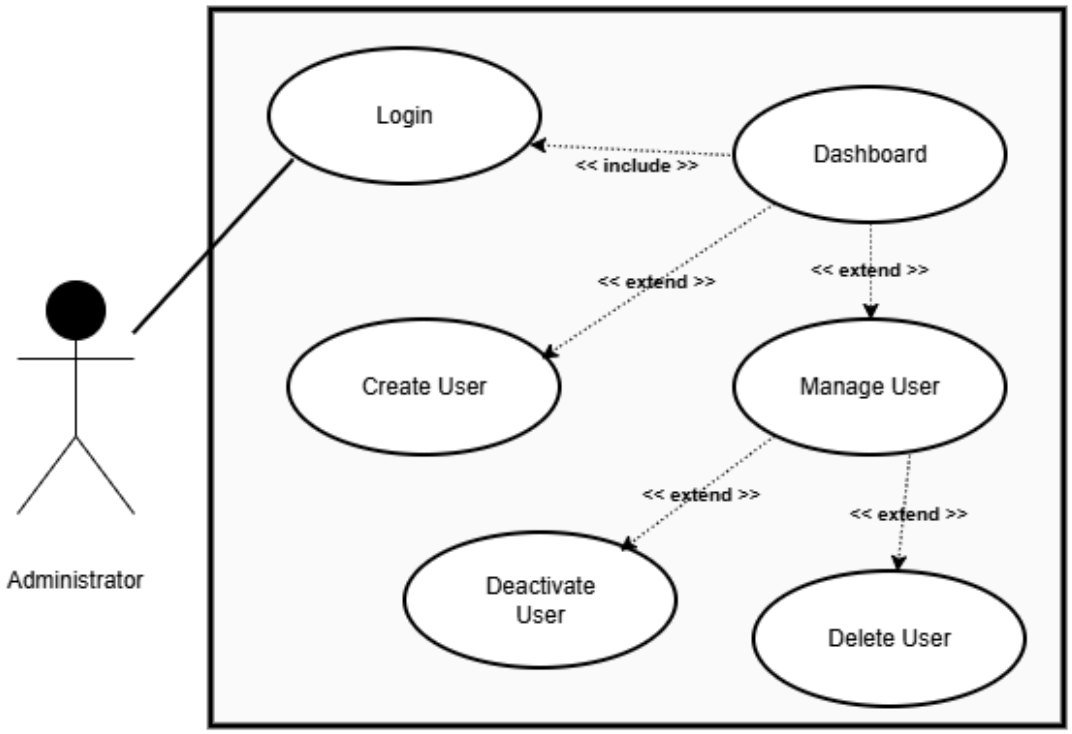



Figure 6. Use Case Diagram of Account Management Module

**Account Management Module.** This part handles all user-related administrative tasks to ensure secure system access. It provides the College Office Administrator with comprehensive control, including the ability to create new user accounts using university emails, assign specific user roles (Administrator, Chairperson, or Faculty staff), and monitor all existing accounts within the system. Additionally, to maintain data integrity, user information is restricted and cannot be changed by the owner of the account once established.

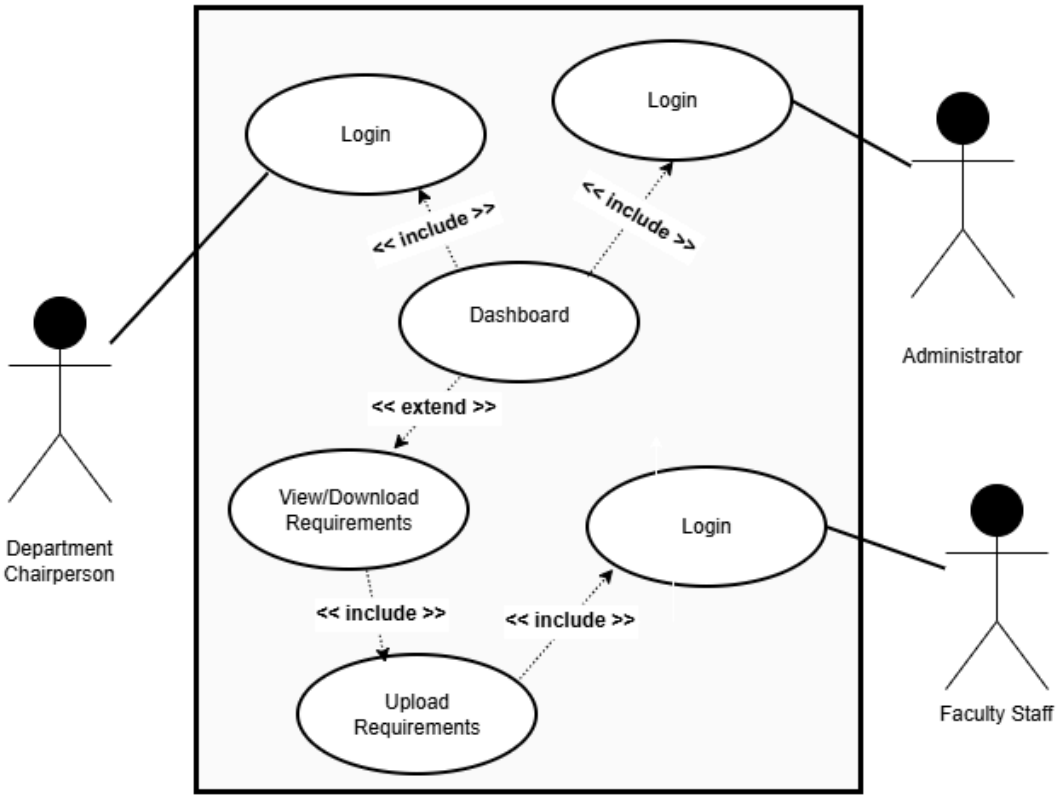



Figure 7. Use Case Diagram of Record / Data Management Module

**Record / Data Management Module.** This module serves as the core operational area for handling academic requirements and is structured hierarchically to support different levels of college authority. The Department Chairperson is granted the capability to view uploaded documents for their respective departments, while the Administration can view all submissions to ensure compliance. Meanwhile, Faculty Staff are provided with the primary function to upload documents, transitioning the college from manual, paper-based submissions to a more efficient digital repository.

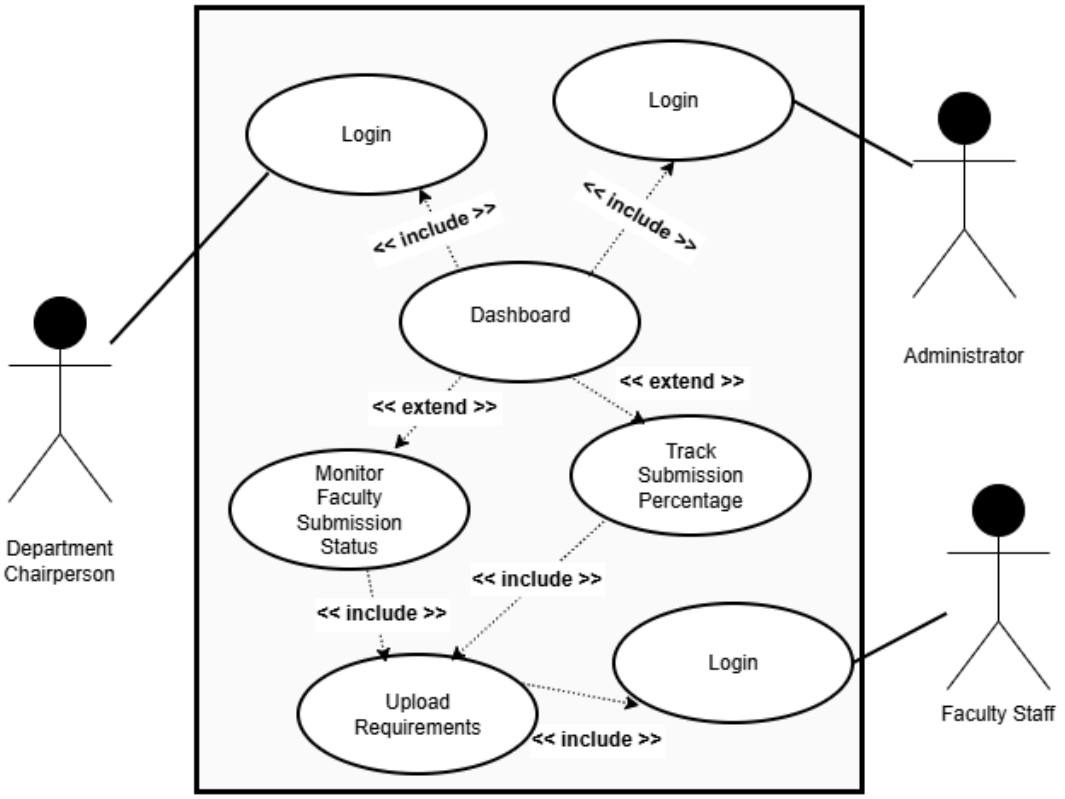



Figure 8. Use Case Diagram of Report / Summary Management Module

**Report / Summary Management Module.** This module provides high-level reporting capabilities for administrative oversight. It enables the College Office Administrator to view all uploaded documents collectively, check the real-time status of specific submissions, and track the overall submission percentage. This ensures that the web-based system can generate accurate compliance summaries for certification purposes in a matter of seconds.

**Non-Functional Requirements**

In the non-functional requirements for the system development., the system must possess the quality of being functional, reliable, usable, efficient, maintainable, and portable.

Table 2. Non-Functional Test Cases for the CollTrace Web-Based College Tracer Management System

| NFR NO. | QUALITY ATTRIBUTES | DESCRIPTION |
|---|---|---|
| NFR01 | Functionality | The system's module and features are all working depending on its specified functionality in the system. |
| NFR02 | Reliability | The system can do its tasks and perform its function in a long period of time without experiencing failures and technical issues. |
| NFR03 | Usability | The system itself is easy to understand for its intended users. |
| NFR04 | Efficiency | The system's modules do not require a high number of resources, and can function even on heavy workloads. |
| NFR05 | Maintainability | The user can easily identify and fix errors or bugs within the system. |
| NFR06 | Portability | The system is compatible and can be accessed through various devices. |

**Budgetary Requirements**

The development of the Web-Based Record Management System for Cavite State University's Extension Office operates within a limited budget constraint. As this is a university-based system development project, there is no dedicated financial allocation for the implementation. The project will leverage existing university resources and infrastructure to minimize costs while maximizing the utilization of available assets.

**Hardware Resources**

The system will utilize existing hardware resources already available within Cavite State University.

Table 3. Budgetary table of Hardware Resources

| Hardware Item | Quantity | Source | Cost |
|---|---|---|---|
| Laptop/PC | As available | Existing university equipment at the Extension Office and campuses | ₱0.00 |
| Server | 1 | University's existing server infrastructure | ₱0.00 |
| Network Infrastructure | As installed | Existing campus network across the main and 12 satellite campuses | ₱0.00 |
| **Total Hardware Cost: ₱0.00** | | | |

**Software Resources**

The system development will utilize a combination of professional hosting services and open-source development tools to ensure high availability and security for the university's records.

Table 4. Budgetary table of Software Resources

| Software Item | Purpose | License Type | Cost |
|---|---|---|---|
| Operating System (Windows/Linux) | Development and deployment platform | Existing university licenses | ₱0.00 |
| Web Browser | System access and testing | Free/Open-source | ₱0.00 |
| Web Hosting Plan (1 Year) | Hosting the web application and database | Commercial Subscription | ₱3000.00 |
| Domain Name | Unique address for accessing the system | Annual Registration | ₱800.00 |
| Development Tools | Coding and debugging | Free/Open-source | ₱0.00 |
| Backup Storage | External backup storage for database and file redundancy | Subscription Service | ₱1000.00 |
| **Total Software Cost: ₱4800.00** | | | |

**Miscellaneous Costs**

Table 5. Budgetary table of Miscellaneous Costs

| Item | Description | Estimated Cost |
|---|---|---|
| Documentation Materials | Printing of documentation papers including drafts and revisions | ₱1000.00 |

| | | |
|---|---|---|
| Presentation Materials | Capstone/System presentation | ₱0.00 |
| Training Materials | User manual and training guides | ₱300.00 |
| Contingency | For unforeseen expenses | ₱500.00 |
| **Total Miscellaneous Cost: ₱1800.00** | | |

**Connectivity Resources**

Internet connectivity will be provided through the existing university infrastructure.

**Total Project Budget Summary**

Table 6. Total Project Budget Summary Table

| Category | Cost |
|---|---|
| Hardware Resources | ₱0.00 |
| Software Resources | ₱4800.00 |
| Miscellaneous Cost | ₱1800.00 |
| Connectivity Resources | ₱0.00 |
| **Total Project Cost: ₱6600.00** | |

| 2026-2027 | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Jan | Feb |
| **Planning** | | | | | | | | | | | | |
| **Documentation** | | | | | | | | | | | | |
| **Paper approval** | | | | | | | | | | | | |

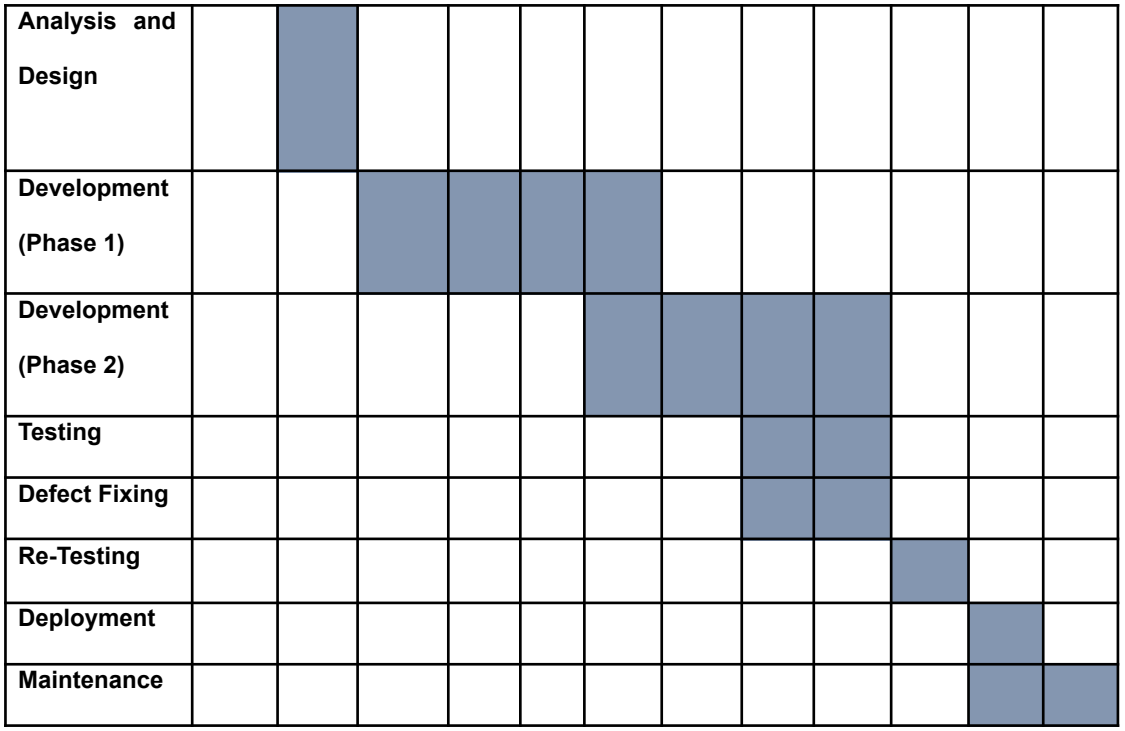



Figure 9: Gantt Chart for School Year 2026-2027

The development and implementation of CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements follows a structured timeline spanning the academic year 2026–2027. The project timeline is designed to ensure systematic development, thorough testing, and successful deployment while aligning with the university's academic calendar and operational requirements. The Gantt chart illustrates the sequential and parallel activities required to complete the project from initial planning through full implementation and maintenance.

**Project Timeline Structure**

The project is divided into distinct phases, each with specific deliverables and milestones. The timeline accounts for the academic schedule, allowing for adequate development time while ensuring minimal disruption to the ongoing operations of the College of Engineering and Information Technology (CEIT).

- **Planning (March 2026)**

  The Planning phase spans one month in March 2026 and serves as the foundation for the entire project. During this critical initial phase, the project team will establish the comprehensive project scope and objectives while identifying all stakeholder requirements to ensure the system meets the needs of the CEIT office, Department Chairpersons, and Faculty Staff. The team will analyze the existing manual processes and Google Drive-based submission workflow to determine the specific functional and non-functional requirements for CollTrace.

- **Documentation (April 2026 – February 2027)**

  The Documentation phase runs parallel to all other development phases from April 2026 to February 2027, ensuring that comprehensive documentation is maintained throughout the entire project lifecycle. This phase includes the process of getting approval, requesting the signing of necessary papers, and conducting surveys and interviews to validate the identified problems within the study. Taking photos of the system's development progress, conducting user surveys, and documenting training sessions are best practices that will be observed to ensure consistency across the development team and facilitate future maintenance of CollTrace.

- **Paper Approval (May 2026 – February 2027)**

  The Paper Approval phase runs from May 2026 to February 2027 and involves securing the necessary institutional approvals required to proceed with the development and deployment of CollTrace. This includes the submission of required documents to the college and university

panels, addressing revisions as needed, and ensuring that all ethical and academic requirements for the conduct of the study are properly fulfilled.

- **Analysis and Design (April 2026 – May 2026)**

  The Analysis and Design phase spans two months from April to May 2026 and represents the critical transition from planning to development. During this phase, the project team will conduct an in-depth analysis of all gathered requirements to ensure a complete understanding of the system's needs and priorities. This includes identifying the specific problems in the current sem-ender requirement submission process of CEIT, determining the feasibility of the proposed solution, and designing the system architecture, database structure, and user interface mockups for CollTrace.

- **Development Phase 1 (May 2026 – August 2026)**

  Development Phase 1 spans four months from May 2026 to August 2026 and focuses on implementing the core functionalities of CollTrace. The development team will begin by setting up the development environment, including the installation and configuration of XAMPP for the local web server, MySQL for database management, and Visual Studio Code as the primary code editor. Core features to be developed in this phase include the login and authentication system, role-based dashboards for the Administrator, Department Chairpersons, and Faculty Staff, and the basic document submission and tracking module.

- **Development Phase 2 (August 2026 – October 2026)**

  Development Phase 2 runs from August 2026 to October 2026 and focuses on implementing the advanced features of CollTrace, including

automated compliance tracking, real-time submission status monitoring with colored status labels, deadline countdown notifications, and the generation of compliance reports per department. This phase also covers the enhancement of user interfaces and the optimization of system performance to ensure a seamless experience for all user roles.

- **Testing (September 2026 – October 2026)**

  The Testing phase spans two months from September to October 2026 and involves a comprehensive and systematic evaluation of CollTrace to identify defects, validate functionality against requirements, and ensure the system is reliable, secure, and ready for deployment. The developers will conduct training sessions for CEIT staff, including Department Chairpersons and Faculty Staff, to allow them to interact with the system and provide feedback as part of the user acceptance testing process.

- **Defect Fixing (September 2026 – October 2026)**

  The Defect Fixing phase runs in parallel with testing from September to October 2026 and is dedicated to systematically resolving all bugs and issues identified during the testing phase. The development team will prioritize defects based on their severity and impact on system functionality, ensuring that CollTrace meets the required quality standards before its final deployment to the CEIT environment.

- **Re-Testing (November 2026)**

  Re-Testing spans November 2026 and represents the final comprehensive validation of CollTrace before production deployment, with emphasis on user acceptance testing and final verification that all

identified defects have been resolved and that all system requirements have been fully met.

- **Deployment (January 2027)**

  The Deployment phase in January 2027 represents the critical transition of CollTrace from development to full production operation within the CEIT environment. The necessary server infrastructure managed by CEIT will be prepared, and the system will be configured for live access by all designated users. Proper coordination with the Dean's Office will be conducted to schedule the deployment and ensure minimal disruption to college operations.

- **Maintenance (February 2027 – onwards)**

  The Maintenance phase begins after the successful deployment of CollTrace and will continue until the developers graduate from the university. During this phase, the team will monitor the system for any new errors or issues reported by users, release patches or updates as necessary, and ensure that the system continues to operate reliably in support of CEIT's sem-ender requirement management processes.

The development and implementation of the Web-Based Record Management System for Cavite State University's Extension Office follows a structured timeline spanning two academic years (2025-2027). The project timeline is designed to ensure systematic development, thorough testing, and successful deployment while aligning with the university's academic calendar and operational requirements. The Gantt charts illustrate the sequential and parallel activities required to complete the project from initial planning through full implementation and maintenance.

**Software Design**

The researchers will provide diagrams that explain the processes of the system to be developed, which is composed of context diagrams and system architecture.

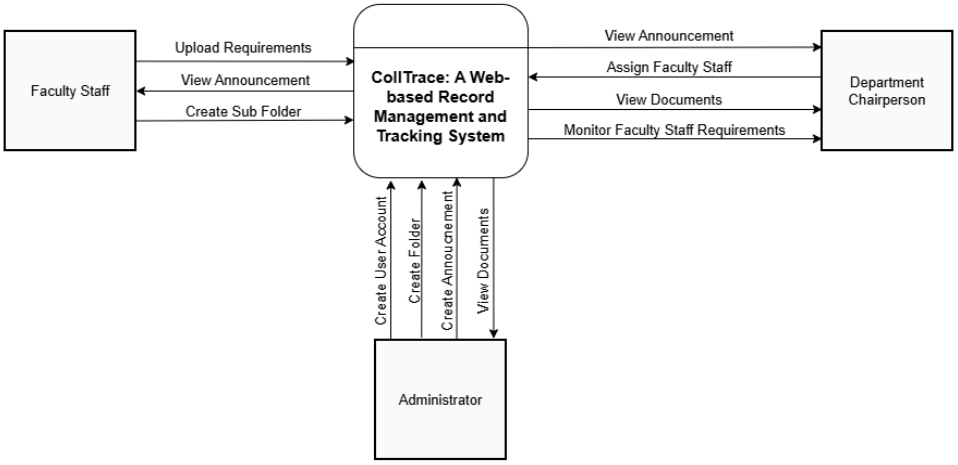



Figure 10. Context Diagram of Web-Based Record Management and Tracking System

The context diagram of a Web-Based Record Management and Tracking System, shown in Figure 10, illustrates the system and its interaction with external entities, particularly the Faculty Staff, Chairperson, and Administrator. The arrows in the diagram represent the flow of data between the users and the system. In this setup, the Faculty Staff provides input to the system in the form of uploaded academic requirement documents. The system then processes this information and stores it in a centralized digital repository.

After processing, the system generates submission statuses and compliance reports, which are then accessed by the Department Chairperson and Administrator for monitoring and for certification purposes. The Administrator, on the other hand, manages user accounts, assigns roles, and ensures proper access to system functions. Through this interaction, the system replaces the manual, google drive

submission process with a more organized and efficient digital workflow, allowing faster document uploading, tracking, and reporting.

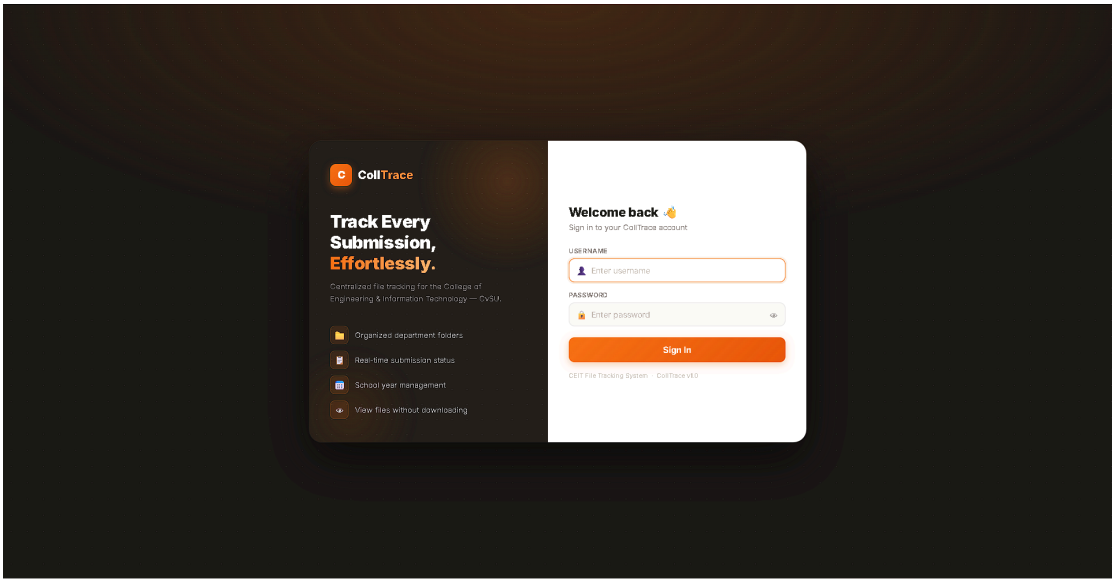



Figure 11. Login Page

This is where users can log in using their existing account credentials. If they do not yet have an account, they need to notify the admin since they are the one who creates the account. This page serves as the main entry point for user authentication.

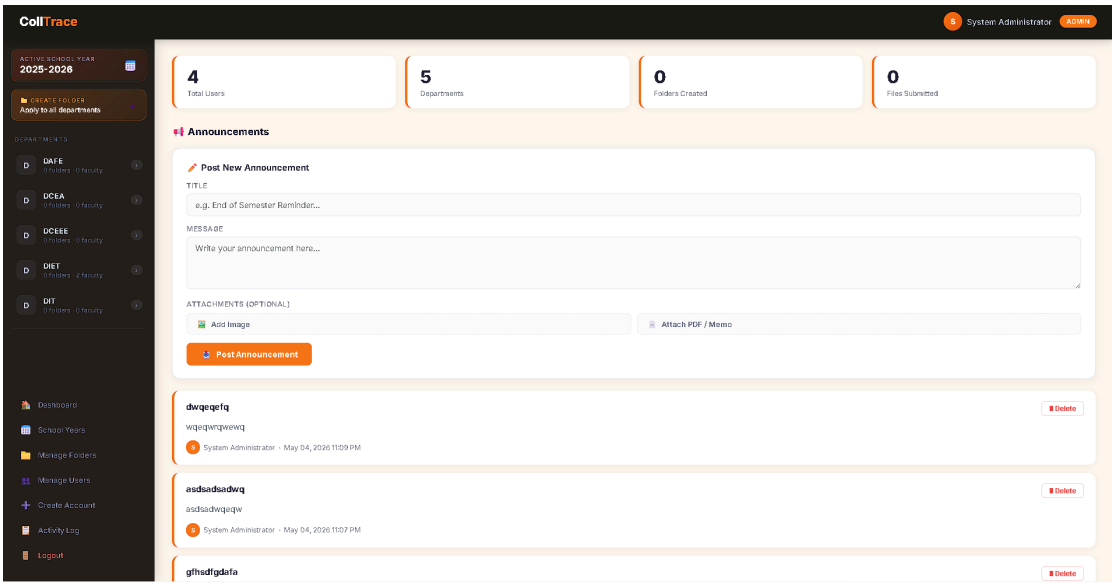



Figure 12. Administrator Dashboard

The administrator's landing page functions as the main dashboard upon login, granting access to all administrative features and system navigation. From this page, the administrator can create and manage user accounts, assign roles, and oversee the system's core functionalities.

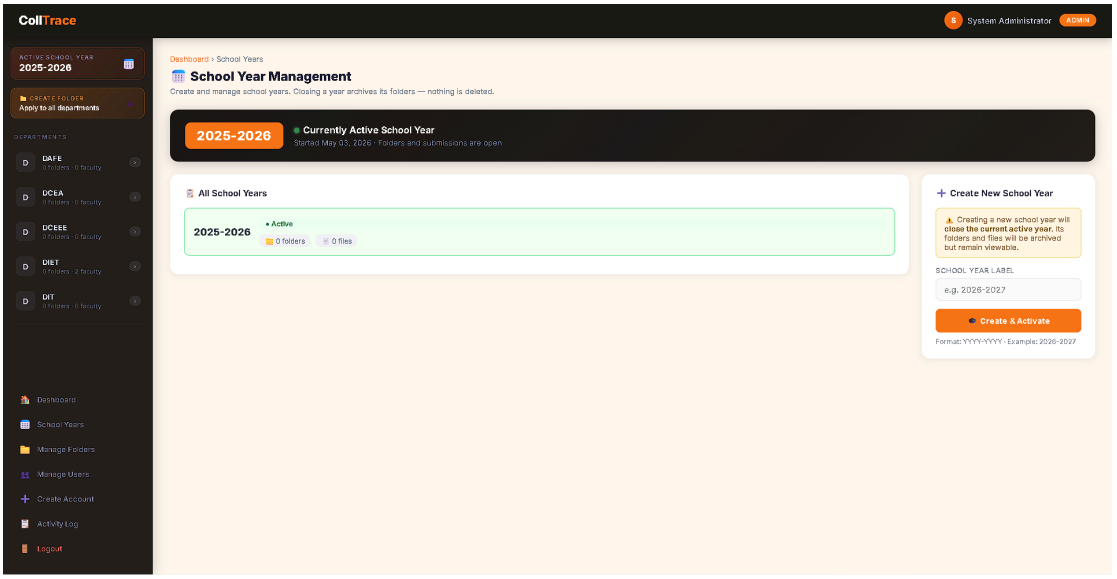



Figure 13. Administrator School Year Management Page

The School Year Management page allows the Administrator to create and manage academic school year folders within the system. For each new academic term, the Administrator can set up a corresponding school year entry, which will serve as the organized directory for all faculty document submissions during that period. This ensures that academic records are systematically arranged and easily retrievable by school year, eliminating the confusion caused by the previous manual folder management process.

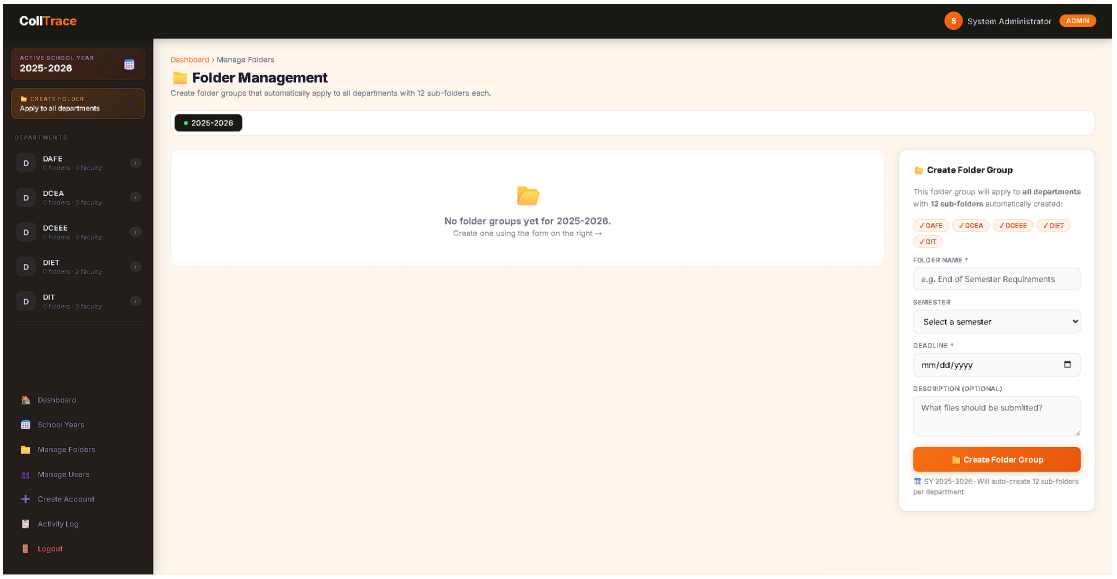



Figure 14. Administrator Folder Management page

The Folder Management page enables the Administrator to create a document submission folder for a selected school year in a single action. Once created, the system will automatically apply and generate the corresponding folders across all five departments simultaneously, each with a designated submission deadline. This streamlines the folder setup process, as the Administrator no longer needs to manually generate Google Drive links and create individual folders per department, ensuring a consistent and organized submission directory for all departments in every academic term.

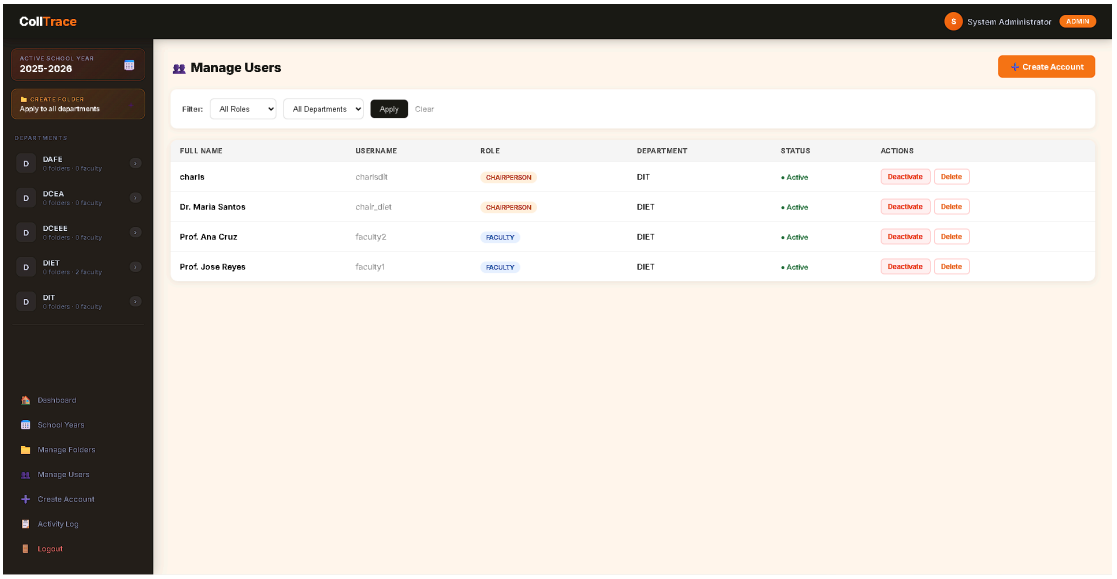


Figure 15. Administrator Manage User Page

The Manage User page enables the Administrator to create and manage user accounts within the system. From this page, the Administrator can register new accounts for Faculty Staff and Chairpersons by assigning their respective university email addresses and user roles. This ensures that all system users are properly authenticated and granted access only to the features corresponding to their designated roles within CollTrace.

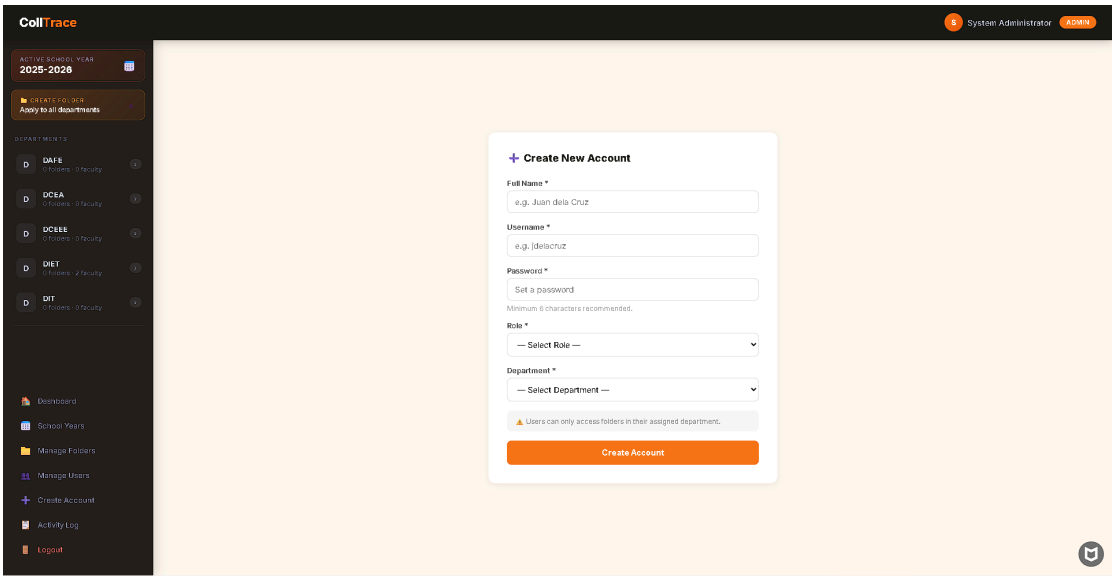



Figure 16. Administrator Create Account Page

The Create Account page allows the Administrator to register new user accounts within the CollTrace system. From this page, the Administrator can input the necessary account details such as the full name, university email address, and designated user role of the new user, whether as a Department Chairperson or Faculty Staff. Once the account is successfully created, the registered user will be able to log in to the system using their assigned credentials. This ensures that only authorized personnel are granted access to the system, maintaining the security and integrity of the CollTrace platform.

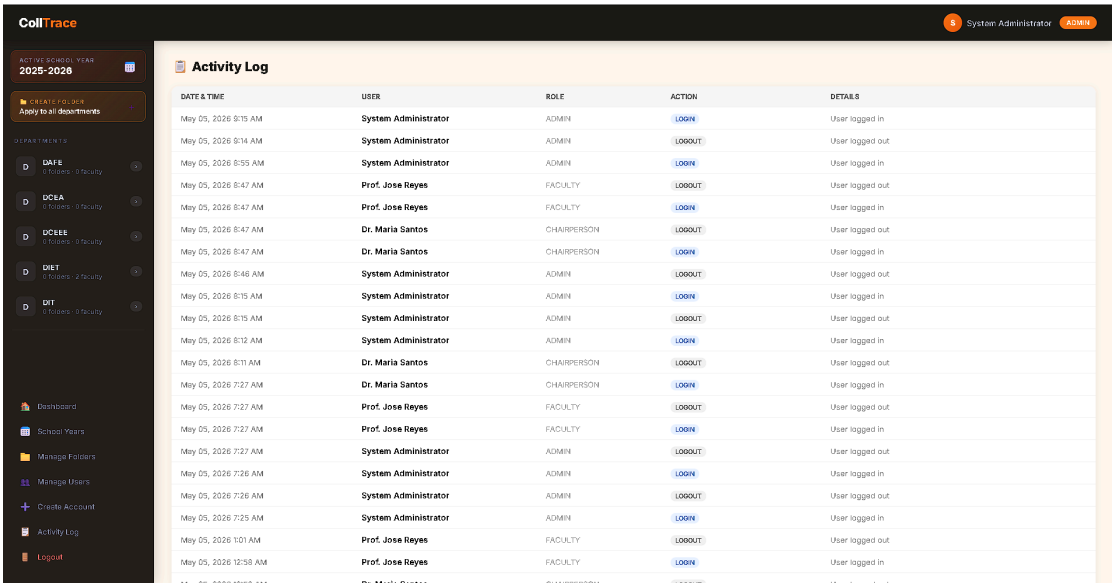



Figure 17. Administrator Activity Log Page

The Activity Log page provides the Administrator with a comprehensive record of all actions and activities performed within the system. This feature allows the Administrator to monitor and track user activities, including account creation, document uploads, and folder management, ensuring transparency and accountability among all system users.

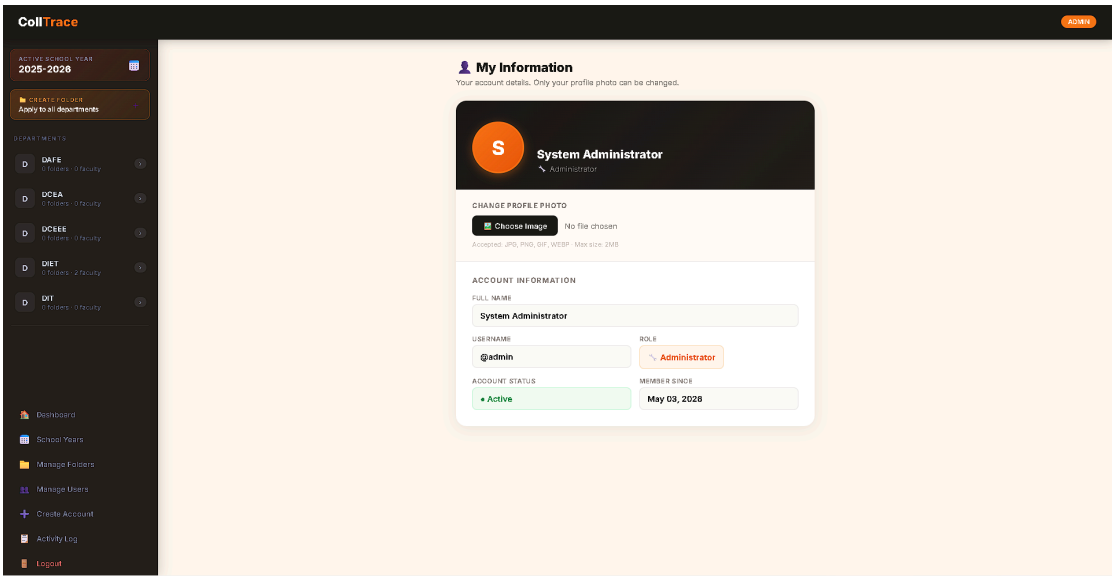



Figure 18. Administrator Profile Management

The Profile Management page allows the Administrator to view and manage their personal account information within the system. From this page, the

Administrator can update their profile picture and view their account details such as their name, assigned role, and university email address associated with their CollTrace account.

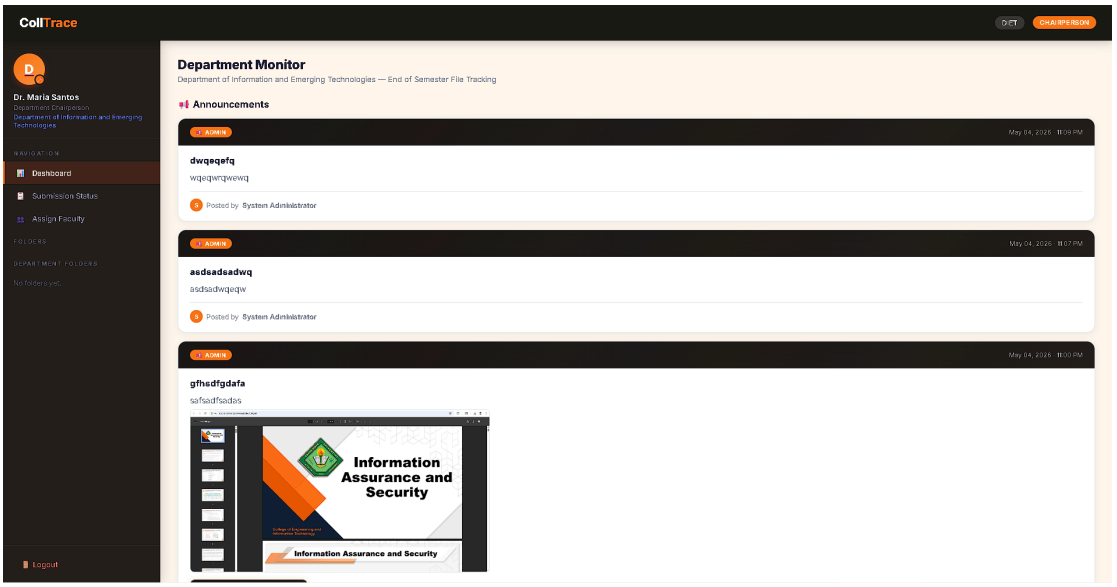



Figure 19. Department Chairperson Dashboard Page

The Chairperson's dashboard serves as the main landing page upon login, providing an overview of the system's key features and functions. From this page, the Chairperson can view announcements posted by the Administrator, ensuring that all department heads are promptly informed of any updates, reminders, or notices relevant to the document submission process within ollTraceCollTrace.

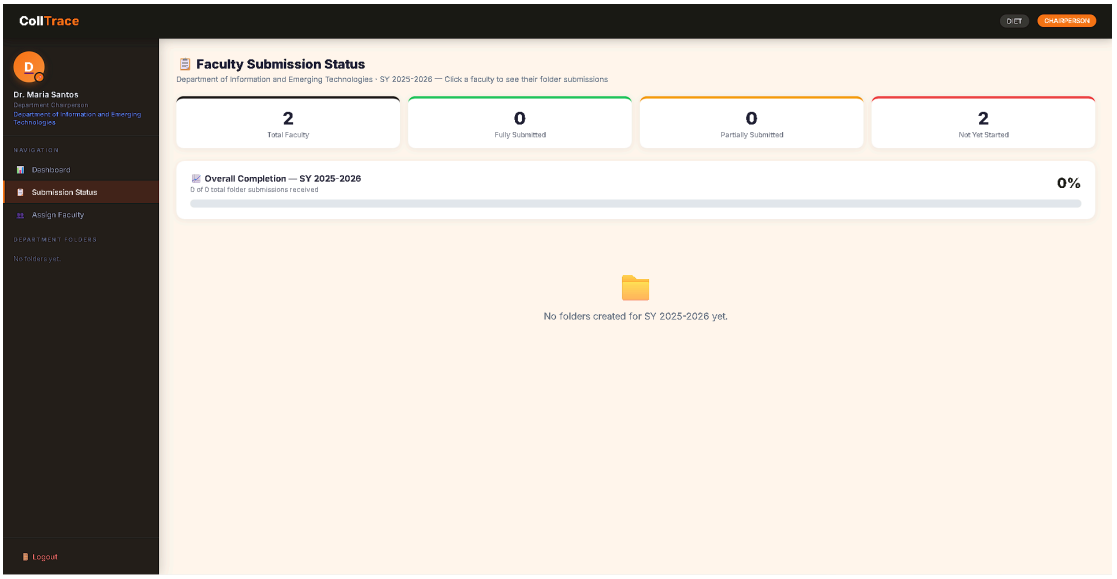



Figure 20. Department Chairperson Submission Status Page

The Submission Status page allows the Chairperson to monitor the document submission progress of all Faculty Staff within their respective department. From this page, the Chairperson can view the current submission status of each faculty member, enabling them to identify who has already complied and who has yet to submit the required academic documents for the ongoing school year.

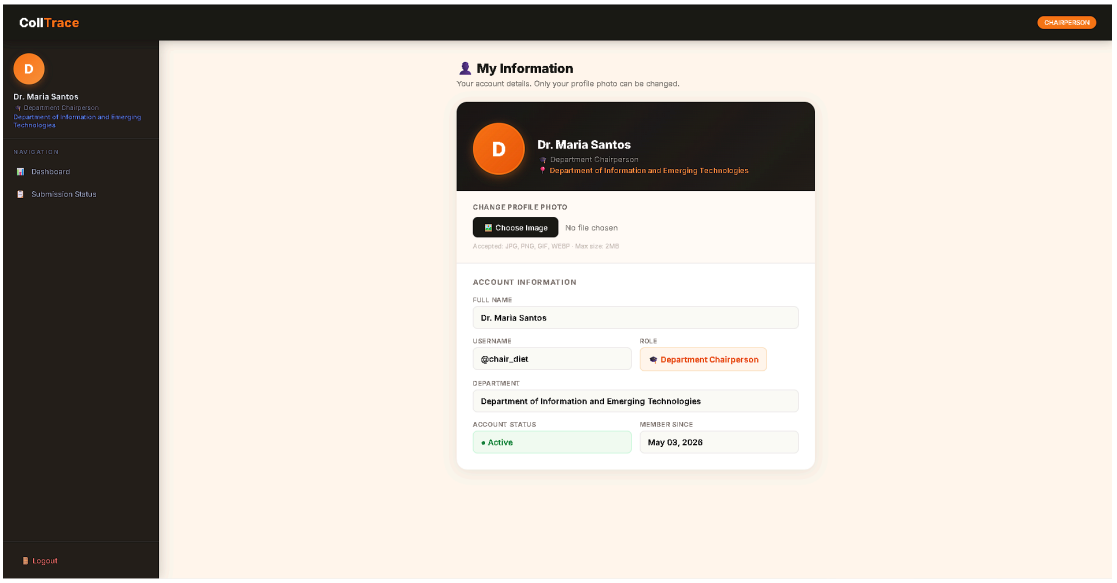



Figure 21. Department Chairperson Profile Management Page

The Profile Management page allows the Chairperson to view their personal account information within the system. From this page, the Chairperson can view their account details such as their name, assigned role, and university email address. Additionally, the Chairperson has the ability to update their profile picture to personalize their CollTrace account.

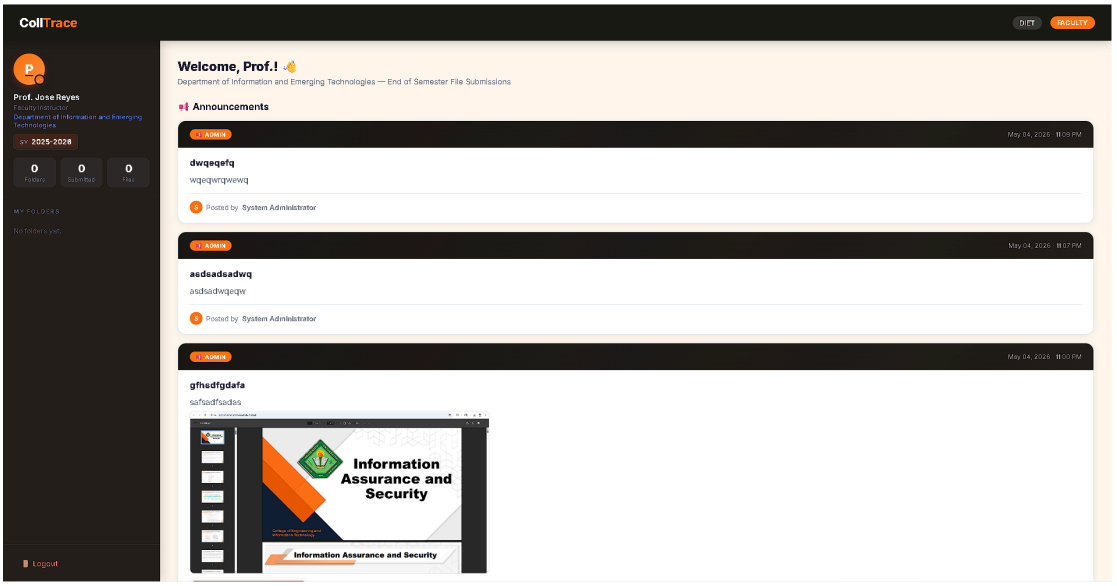



Figure 22. Faculty Staff Dashboard Page

The Faculty Staff dashboard serves as the main landing page upon login, providing an overview of the system's key features and functions. From this page, the Faculty Staff can view announcements posted by the Administrator, ensuring that they are promptly informed of any updates, reminders, or notices relevant to the document submission process within CollTrace.

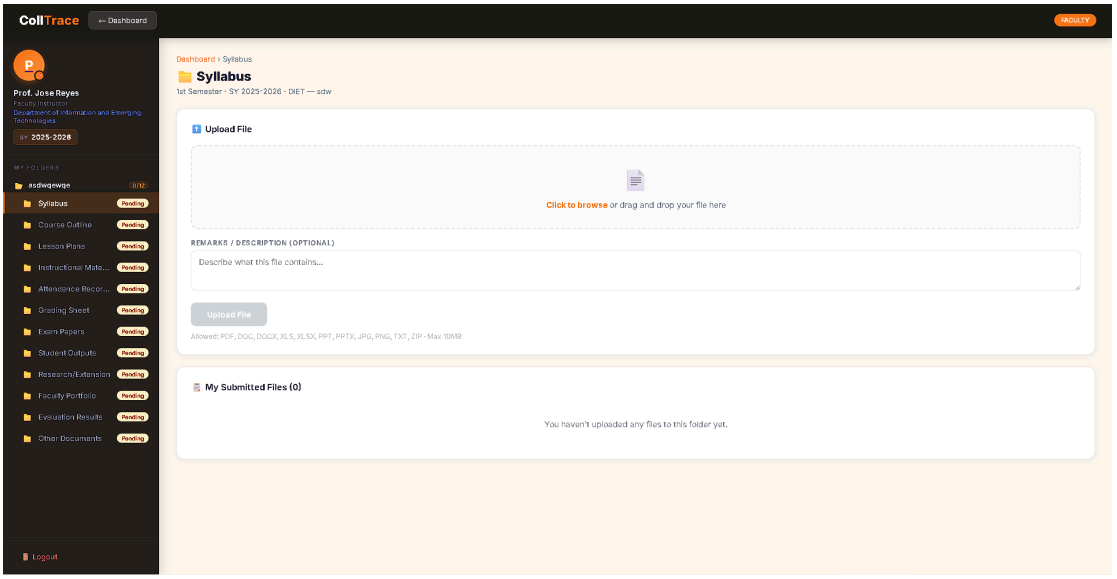



Figure 23. Faculty Staff Folder Management Page

The Folder Management page serves as the primary submission interface for Faculty Staff within CollTrace. From this page, faculty members can access the designated folders for their respective department and upload the required academic documents for the current school year. This ensures that all submissions are properly organized and stored within the correct directory, replacing the previous process of manually uploading files to shared Google Drive folders that were prone to accidental deletion and limited storage capacity.

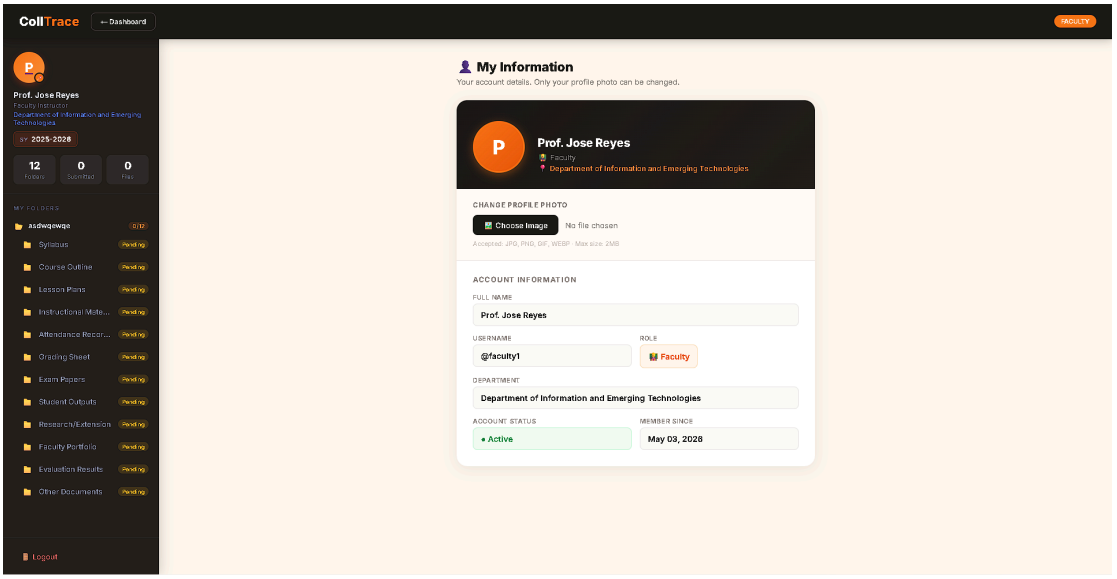



Figure 24. Faculty Staff Profile Management Page

The Profile Management page allows the Faculty Staff to view their personal account information within the system. From this page, the Faculty Staff can view their account details such as their name, assigned role, and university email address. Additionally, the Faculty Staff has the ability to update their profile picture to personalize their CollTrace account.

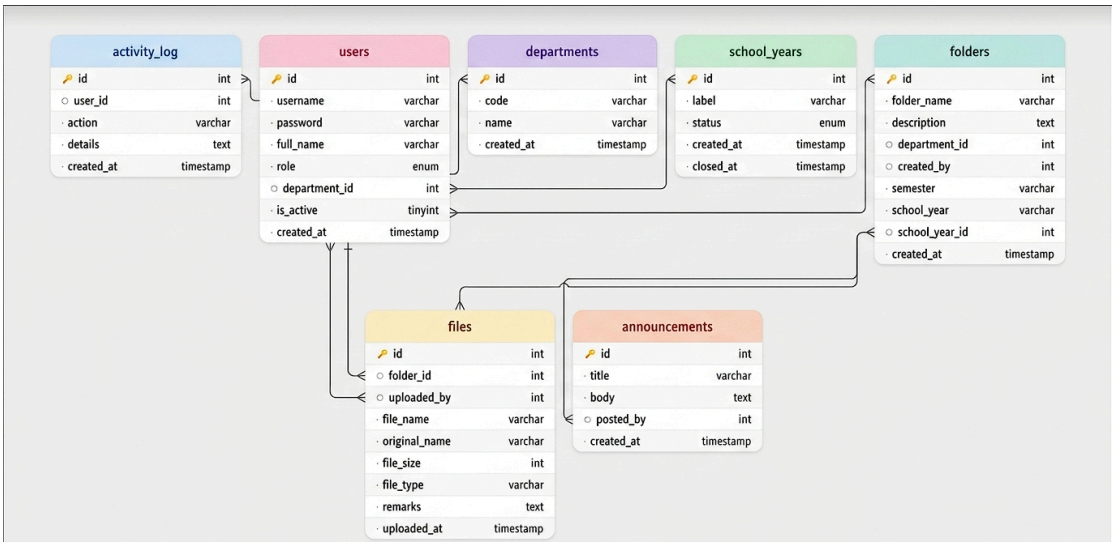



Figure 25. Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) serves as the structural foundation of the CollTrace: A Web-Based College Tracer Management System, ensuring a well-defined relational database structure. The system enforces foreign key constraints to maintain referential integrity and consistency across all data. The database schema is normalized, and its entities are specifically designed to support the three core functional modules of the system:

**a. Account Management Module:** This module is supported by the users and activity_log tables. The users table implements Role-Based Access Control (RBAC) by defining user roles such as Administrator, Chairperson, and Faculty Staff. Each user is also linked to a specific department through the department_id attribute. A one-to-many relationship exists between users and activity_log, ensuring that all system activities—such as account creation, file uploads, and folder management—are properly recorded and traced to a specific user. This guarantees accountability, security, and transparency within the system.

**b. Record/Data Management Module:** This module is built upon the folders, files, departments, and school_years tables. The folders table organizes submission directories based on academic year and semester, while the files table stores all uploaded faculty requirements. A one-to-many relationship between folders and files allows multiple documents to be stored under a single structured directory. The departments and school_years tables serve as reference entities to ensure that all records are properly categorized and organized. This design eliminates redundancy and provides a centralized repository for managing academic documents efficiently.

**c. Report/Summary Management Module:** This module utilizes the relationships across files, folders, users, and school_years to generate real-time submission tracking and compliance summaries. By linking uploaded documents to specific faculty members and academic periods, the system can compute submission

statuses and overall compliance percentages. These aggregated data outputs are presented to administrators and chairpersons, enabling efficient monitoring, evaluation, and reporting of faculty submission performance.

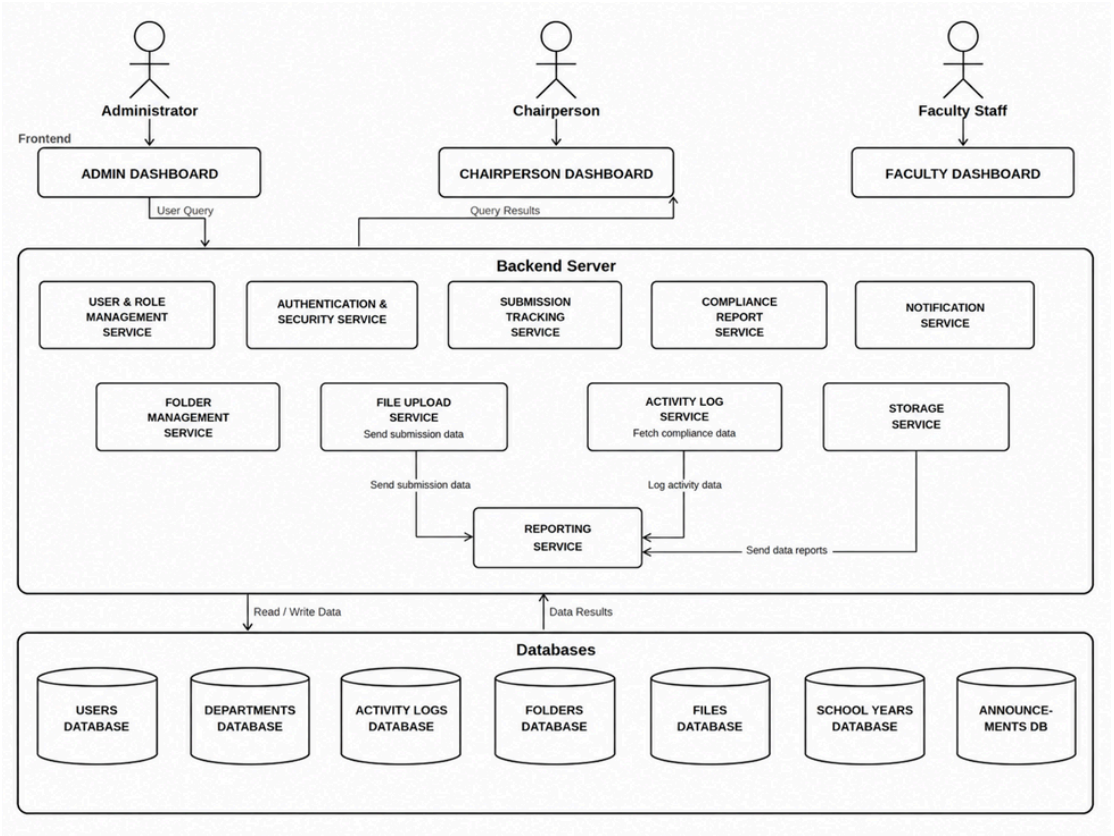



Figure 26. System Architecture Diagram

The system's internal structure is built using a three-tier architecture, as shown in Figure 26. This setup separates the different parts of the system to ensure it stays secure and processes data quickly.

The Frontend provides specific interactive screens: an Admin Dashboard for the Administrator, a Chairperson Dashboard for the Department Chairperson, and a Faculty Dashboard for the Faculty Staff. This ensures each user only accesses the tools meant for their specific role. When users interact with these dashboards, they send a "User Query" down to the server, and the system sends back "Query Results" to display the requested information.

These queries are processed by the Backend Server, which acts as the brain of the system. As shown in the diagram, its internal services actively communicate with each other to complete tasks:

a. **Authentication & Security Service:** Checks logins and passwords to keep the system secure and ensures only authorized users can access the system.

b. **User & Role Management Service:** Controls user accounts and defines what each user — Administrator, Department Chairperson, or Faculty Staff — is allowed to do inside the system.

c. **Submission Tracking Service:** Monitors the real-time submission status of faculty requirements per department, and sends submission data directly to the Reporting Service for compliance monitoring.

d. **Compliance Report Service:** Generates percentage-based compliance summaries per department by fetching compliance data and forwarding it to the Reporting Service.

e. **Notification Service:** Automatically alerts faculty members who have not yet submitted their required academic documents before the deadline.

f. **Folder Management Service:** Handles the creation and organization of academic year folders and department subfolders used for document submission.

g. **File Upload Service:** Manages the secure uploading and storage of faculty academic requirement documents into their designated folders.

h. **Activity Log Service:** Records all system actions such as account creation, file uploads, and folder management to maintain accountability and transparency.

i. **Storage Service:** Safely handles the physical storage of uploaded documents and sends data reports to the Reporting Service.

j. **Reporting Service:** Creates compliance summaries by fetching submission data from the Submission Tracking Service, compliance data from the Compliance Report Service, and data reports from the Storage Service.

Finally, the backend safely stores all system records by continuously sending “Read / Write Data” commands and receiving “Data Results” from the Databases layer. To keep information highly organized, this storage is divided into specific collections: Users Database, Departments Database, Activity Logs Database, Folders Database, Files Database, School Years Database, and the Announcements Database.

**System Development**

The development of a Web-Based Record Management and Tracking System will follow a structured set of procedures based on the Agile Software Development Life Cycle. The researchers will begin by translating the identified requirements into system design and functional modules. A database will be designed, where tables and relationships will be created to store user accounts, academic requirement records, and submission data. The system interface will be developed with a focus on usability and ease of navigation for the Administrator, Department Chairperson, and Faculty Staff. The core functionalities, which will include account management, record and data management, and report and summary management, will subsequently be implemented. Continuous testing and debugging will be conducted throughout each iteration to ensure that errors will be identified and corrected at the earliest stage possible. The system will then undergo final testing and refinement prior to its deployment.

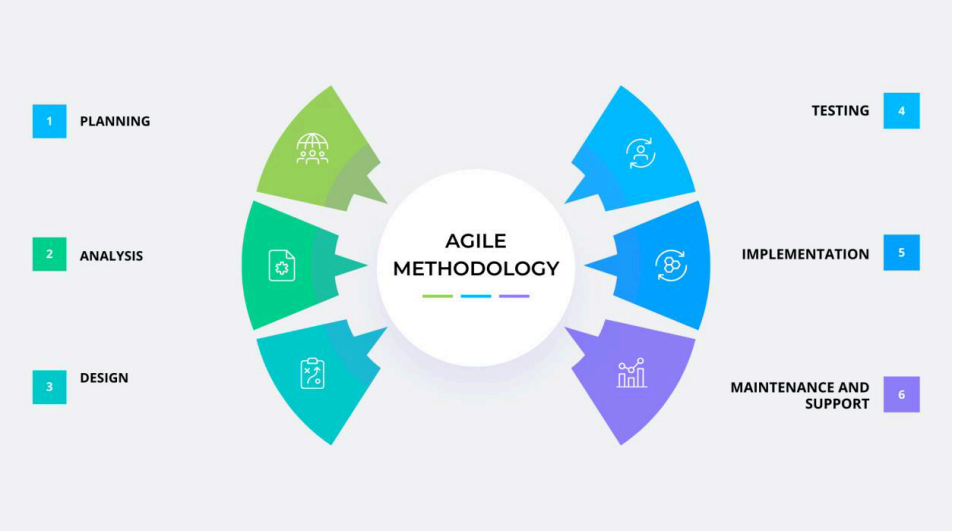



Figure 27. Agile Development Methodology (Yas et al., 2023)

**Planning Phase**

According to Yas et al. (2023), the requirements analysis phase is the first and one of the most important stages in the software development process. In this phase, a dedicated team gathers and documents the needs and expectations of the customer in a clear and organized manner. The collected information is then compiled into a Software Requirements Specification (SRS) document, which serves as the foundation for the entire development process. Based on this document, the developers create a project plan and evaluate whether the project is technically and economically feasible (p. 175).

The development operation began with the planning phase, where the developers sought requirements through interviews, surveys, and consultations with the College Office Administrator, Department Chairpersons, and Faculty Staff of the College of Engineering and Information Technology (CEIT). The specific requirements, difficulties, and expectations of the end users in relation to the web-based record management and tracking system were determined at this phase.

**Analysis Phase**

Yas et al. (2023) explained that the analysis phase comes after the initial requirements have been gathered. During this stage, the collected information is carefully reviewed and organized to better understand what the system needs to do. The goal is to clearly identify the features and functions that the system must have in order to meet the needs of its users, which then guides the next stages of development (p. 175).

After gathering the requirements, the developers proceeded to the analysis stage, where they reviewed the collected data and determined the features and overall structure of the system. This included identifying the specific needs of each user role — namely the Administrator, Chairperson, and Faculty Staff — in relation to document submission tracking and folder management within CEIT.

**Design Phase**

According to Yas et al. (2023), the design phase is where the requirements gathered and analyzed in the previous stages are translated into a concrete plan for building the system. This includes deciding on the system's overall structure, how it will look, and how its different parts will work together. The design is recorded in a Software Design Description (SDD) document, which is reviewed by experts to check its quality, feasibility, and alignment with the project goals before development begins (p. 175).

Subsequently, the developers created the system architecture, database structure, and user interface designs based on the analyzed requirements. This phase also involved selecting the appropriate frameworks, technologies, and tools for system development, including PHP, MySQL, HTML, and CSS. Before beginning

actual development, clients were shown the design prototypes in order to gather their feedback and ensure that the proposed design satisfies their needs.

**Testing Phase**

As stated by Yas et al. (2023), the testing phase is carried out to find and fix errors in the system before it is released to users. Since individual unit testing may not always catch all issues, a complete system test is also performed to check how all parts of the system work together. This phase is important for ensuring that the system functions correctly and meets the expected standards of quality (p. 175).

Regular meetings and demonstrations with stakeholders were held throughout the development process to update them on progress, gather their feedback, and make any required adjustments. Testing was conducted regularly during each sprint to identify and resolve issues at the earliest stage possible, ensuring the system's overall reliability and quality.

**Implementation Phase**

Yas et al. (2023) described the implementation phase as the stage where the actual coding of the system takes place, based on the design specifications that were previously prepared. Each part of the system is built and tested individually to ensure that it works as intended, and any necessary corrections are made along the way before moving on to the next component (p. 175).

Before being deployed and implemented at Cavite State University's College of Engineering and Information Technology, the system underwent final testing and

assessment after all sprints were completed. This ensured that the final system aligned with the actual requirements of the college office and its users.

**Maintenance and Support Phase**

According to Yas et al. (2023), the maintenance phase is the final stage of the software development process, occurring after the system has been delivered and is already in use. During this phase, any remaining errors are identified and corrected, and improvements are made based on real-world usage. The system is also updated over time to ensure that it continues to meet the needs of its users effectively (p. 175).

In order to keep the system current and functional, the Agile methodology also allows the addition of new features or modifications depending on the feedback received from the Administrator, Chairpersons, and Faculty Staff of the proposed CollTrace system.

The system will be developed using PHP as the server-side programming language and MySQL for database management. For the front-end, HTML, CSS, and JavaScript will be utilized to create a responsive and user-friendly interface. XAMPP will serve as the local development environment, while Visual Studio Code will be used for coding and editing. Additional tools such as draw.io will be used for diagram creation and documentation.

The hardware requirements for the development of the system will include a 64-bit architecture desktop or laptop with a minimum of 4GB system memory, at least 20GB of available hard drive space, and a minimum of 4 CPU cores. The system will

run on a Windows operating system with a modern web browser such as Google Chrome for testing and execution. No specialized hardware such as biometric devices, RFID, or cameras will be required for the proposed system, as CollTrace is a web-based record management and tracking system. The system will operate within a local or institutional network, requiring a stable internet connection for access and data synchronization. Since the system is web-based, it will also be accessible using mobile devices through a browser, although it will be primarily optimized for desktop use.

**System Testing**

Module Name: CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements

Created / Conducted by: Researchers

Date Created: May 5, 2026

Date Conducted:

**Administrator Test Cases**

Table 7. Administrator test case table

| TEST CASE ID | TEST SCENARIO | TEST CASE | PRECONDITION | TEST STEPS | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|---|---|---|
| TC-A001 | Verify user account creation | Input valid user account details | 1. System is running<br>2. Admin is logged in | 1. Open Manage User page<br>2. Click Create Account<br>3. Input full name, | Valid name, university email, role: Faculty Staff | Account is created successfully and appears in user list | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | universit y email, and role<br><br>4. Click Submit | | | | |
| TC-A00 2 | Verify user accoun t creatio n with missin g fields | Input incomplet e account details | 1. System is running<br><br>2. Admin is logged in | 1. Open Create Account page<br><br>2. Leave some fields blank<br><br>3. Click Submit | Missing email or role field | System prevents submission and shows error message | | |
| TC-A00 3 | Verify folder creatio n for a school year | Create a new folder for a school year | 1. System is running<br><br>2. Admin is logged in | 1. Open Folder Manage ment page<br><br>2. Select school year<br><br>3. Create a folder with deadline<br><br>4. Click Submit | Valid school year, folder name, deadlin e date | Folder is created across all 5 department s simultaneou sly | | |
| TC-A00 4 | Verify school year manag ement | Create a new academic school year entry | 1. System is running<br><br>2. Admin is logged in | 1. Open School Year Manage ment page<br><br>2. Input new school year<br><br>3. Click Save | Valid school year entry (e.g., 2025-2 026) | New school year entry is saved and visible in the list | | |
| TC-A00 5 | Verify activity log visibilit y | View activity log of all user actions | 1. System is running<br><br>2. Admin is logged in | 1. Navigate to Activity Log page<br><br>2. View logged activities | Existing user activity records | All user actions (uploads, account creation, folder manageme nt) are displayed with timestamps | | |

**Department Chairperson Test Cases**

Table 8. Department Chairperson test case table

| TEST CASE ID | TEST SCENARIO | TEST CASE | PRECONDITION | TEST STEPS | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|---|---|---|
| TC-C001 | Verify submission status monitoring | View faculty submission statuses for the department | 1. System is running<br>2. Chairperson is logged in | 1. Open Submission Status page<br>2. View list of faculty members | Existing faculty submission records | System displays each faculty member's submission status with correct color indicators (Yellow/Green/Red) | | |
| TC-C002 | Verify document viewing for the department | View uploaded documents from faculty staff | 1. System is running<br>2.Chairperson is logged in | 1.Navigate to Record/ Data Management<br>2. Select a faculty member<br>3. Open submitted document | Existing uploaded faculty documents | Chairperson can view documents specific to their department only | | |
| TC-C003 | Verify that chairperson cannot view other department records | Attempt to access another department's submissions | 1. System is running<br>2.Chairperson is logged in | 1. Navigate to Record/ Data Management<br>2.Attempt to access a different department's folder | Department access credentials | System restricts access and only shows records from the Chairperson's department | | |
| TC-C004 | Verify announcement visibility | View announcements from Administrator | 1. System is running<br>2.Chairperson is logged in | 1.Navigate to Dashboard<br>2. View announcements section | Existing admin-posted announcements | Announcements posted by the Administrator are displayed correctly on the dashboard | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| TC-C005 | Verify assignment of faculty member to upload a specific requirement | Assign a faculty member to upload a specific requirement | 1. System is running<br><br>2.Chairperson is logged in 3. Faculty members exist within the department | 1. Open Submission Status page<br><br>2. Select a specific requirement slot<br><br>3.Assign a faculty member from department<br><br>4. Click Confirm | Existing faculty member within the same department; specific requirement (e.g., syllabus) | Selected faculty member is successfully assigned to the requirement and is now enabled to upload for that slot | | |

**Faculty Staff Test Cases**

Table 9. Faculty Staff test case table

| TEST CASE ID | TEST SCENARIO | TEST CASE | PRECONDITION | TEST STEPS | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|---|---|---|
| TC-F001 | Verify document upload | Upload a required academic requirement document | 1. System is running<br><br>2. Faculty Staff is logged in | 1. Open Folder Management page<br><br>2. Select the correct folder<br><br>3.Upload document 4. Click Submit | Valid academic requirement file (e.g., TOS, exam) | Document is uploaded successfully and saved in the correct folder | | |
| TC-F002 | Verify upload with invalid file type | Attempt to upload a non-supported file type | 1. System is running<br><br>2. Faculty Staff is logged in | 1. Open Folder Management page<br><br>2.Select a folder<br><br>3.Attempt to upload unsupported file type | File with unsupported format (e.g., .exe) | System rejects the file and displays an error message | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| TC-F003 | Verify submission status indicators | Check own submission compliance status | 1. System is running<br><br>2. Faculty Staff is logged in | 1. Open Folder Management page<br><br>2. Observe status color labels | Existing submission records | Color indicators correctly reflect submission status: Yellow (not yet submitted), Green (submitted), Red (late) | | |
| TC-F004 | Verify announcement visibility | View announcements from Administrator | 1. System is running<br><br>2. Faculty Staff is logged in | 1.Navigate to Dashboard<br><br>2. View announcements section | Existing admin-posted announcements | Announcements posted by the Administrator are displayed correctly on the dashboard | | |
| TC-F005 | Verify profile management | View profile account details | 1. System is running<br><br>2. Faculty Staff is logged in | 1.Navigate to Profile Management page<br><br>2. View account details | Existing account information | Faculty Staff profile displays correct name, role, and university email address | | |
| TC-F006 | Verify that assigned faculty can delegate upload to another faculty within the department | Assigned faculty selects another faculty member within the department to upload on their behalf | 1. System is running<br><br>2. Faculty Staff is logged in<br><br>3. A faculty member has been assigned a requirement by the Chairperson<br><br>4. Other faculty members exist within the same department | 1. Open the assigned requirement slot<br><br>2. Select another faculty member from the department<br><br>3. Delegate upload responsibility<br><br>4. Click Confirm | List of faculty members within the same department | Selected faculty member is designated as the uploader for that requirement slot; original assignee no longer needs to upload the same document | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| TC-F007 | Verify that faculty cannot delegate to a member outside their department | Assigned faculty attempts to select a faculty member from another department | 1. System is running<br><br>2. Faculty Staff is logged in<br><br>3. A faculty member has been assigned a requirement<br><br>4. Faculty members from other departments exist in the system | 1. Open the assigned requirement slot<br><br>2.Attempt to select a faculty member outside the department | Faculty member from a different department | System restricts the selection to members within the same department only; out-of-department members are not shown or selectable | | |
| TC-F008 | Verify that only the delegated faculty member can upload after delegation | Original assignee attempts to upload after delegating to another faculty | 1. System is running<br><br>2. A requirement has been delegated to another faculty member | 1.Original assignee logs in<br><br>2.Attempts to upload to the delegated requirement slot | Delegated requirement slot | System prevents the original assignee from uploading; only the newly designated faculty member retains upload access for that requirement | | |

The test cases shown in Table  will be used to validate the functionality of the CollTrace system. Similar test cases will be created for additional modules of the system as development progresses.

To perform the reliability, accuracy, compatibility, and performance tests, the following procedures will be conducted:

Testing Procedures and Computation of Results

To perform the reliability test, the following procedures will be conducted:

1. Run the system.
2. Allow multiple users to perform document uploads and submissions using sample data.

3. Repeat the submission process several times.
4. Record any system failures or inconsistencies.
5. Tally the results and determine the failure rate.

To perform the accuracy test, the following procedures will be conducted:

6. Compare system-generated submission statuses and compliance reports with expected outputs.
7. Count the number of correct outputs.
8. Compute the accuracy using the formula:

   Accuracy = (Number of Correct Outputs / Total Number of Test Cases) × 100

To perform the compatibility test, the following procedures will be conducted:

9. Run the system in different web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
10. Observe and record system behavior in each browser.
11. Test the system on different operating systems such as Windows and mobile browsers.
12. Record any inconsistencies or issues.

To perform the performance efficiency test, the following procedures will be conducted:

13. Run the system multiple times.
14. Perform document uploads and report generation.
15. Record the response time for each operation.
16. Compute the average response time to evaluate system performance.

All results of the tests will be documented, and actual results, status, and execution details will be filled during the testing phase.

**System Evaluation**

The system evaluation will be conducted to assess the overall quality and acceptability of CollTrace: A Web-Based College Tracer Management System for Sem-Ender Requirements. The evaluation will involve both technical and non-technical evaluators. The technical evaluators will consist of three to five IT experts, such as Information Technology faculty members or industry practitioners, while the non-technical evaluators will include end-users from the College of Engineering and Information Technology (CEIT), specifically the College Office Administrator (1), Department Chairpersons (3), and Faculty Staff (6), who will serve as the primary users of the system.

The system will be evaluated based on the ISO 25010 software quality standards, which include functionality, reliability, usability, efficiency, maintainability, and portability. A structured evaluation questionnaire will be used as the evaluation instrument.

A five-point Likert scale will be used, where 5 represents Excellent, 4 Very Good, 3 Good, 2 Fair, and 1 Poor. The responses will be analyzed using the mean to determine the overall performance of the system. The mean will be computed by dividing the total score by the number of responses. The results will be interpreted to determine the level of system acceptability.

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

**Name of Evaluator:**_____ **Position:**_____

**Instruction:** Please evaluate the software material by using the given scale and placing a checkmark (✓) under the corresponding numerical rating:

| Numerical Rating | Equivalent |
|---|---|
| 5 | Excellent |
| 4 | Very Good |
| 3 | Good |
| 2 | Fair |
| 1 | Poor |

| INDICATORS | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **A. Functionality** | | | | | |
| 1. Suitability (Degree to which the system provides functions that meet specified tasks and user needs) | | | | | |
| 2. Accuracy (Degree to which the system provides correct and precise results) | | | | | |
| 3. Interoperability (Degree to which the system can interact with other systems if needed) | | | | | |
| 4. Security (Degree to which the system protects data from unauthorized access) | | | | | |
| 5. Compliance (Degree to which the system adheres to standards and requirements) | | | | | |
| **B. Reliability** | | | | | |
| 1. Maturity (Degree to which the system performs consistently under normal conditions) | | | | | |
| 2. Fault Tolerance (Degree to which the system continues to operate despite errors) | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| 3. Recoverability (Degree to which the system can recover data after failure) | | | | | |
| **C. Usability** | | | | | |
| 1. Understandability (Degree to which users can understand how to use the system) | | | | | |
| 2. Learnability (Degree to which users can easily learn how to use the system) | | | | | |
| 3. Operability (Degree to which the system is easy to use and control) | | | | | |
| 4. Attractiveness (Degree to which the interface is visually appealing) | | | | | |
| **D. Efficiency** | | | | | |
| 1. Time Behavior (Degree to which response time and processing speed meet requirements) | | | | | |
| 2. Resource Utilization (Degree to which the system uses resources efficiently) | | | | | |
| **E. Maintainability** | | | | | |
| 1. Analyzability (Degree to which system issues can be diagnosed easily) | | | | | |
| 2. Changeability (Degree to which the system can be modified easily) | | | | | |
| 3. Stability (Degree to which the system avoids unexpected failures after changes) | | | | | |
| 4. Testability (Degree to which the system can be tested effectively) | | | | | |
| **F. Portability** | | | | | |
| 1. Adaptability (Degree to which the system can run in different environments) | | | | | |
| 2. Installability (Degree to which the system can be installed easily) | | | | | |
| 3. Co-existence (Degree to which the system can operate alongside other systems) | | | | | |
| 4. Replaceability (Degree to which the system can replace another system for the same purpose) | | | | | |

**System Implementation**

After the system has been completely developed, to ensure the system is properly integrated into the CEIT workflow, the following activities will be conducted:

1. **Pre-deployment Preparation:**
    a. Confirm completion of development and testing phases.
    b. Finalize software documentation, including user manuals for Administrators, Department Chairpersons and Faculty Staffs, installation guides, and release notes.
    c. Conduct a final code review and ensure all issues and bugs are resolved.

2. **Infrastructure set-up**
    a. Set up the necessary software infrastructure for hosting the web-based application, including servers, databases, and other infrastructure handled by the CEIT.
    b. Install and configure required software dependencies and libraries in order to support the existing current version of PHP and MySQL.

3. **Data migration**
    a. Plan and execute data migration activities, including transferring existing data of current Semester Requirements of Faculty Staffs to the new system and creating user accounts for all users.
    b. Develop and execute rollback procedures in case of data migration failures to prevent the loss of important requirements submitted by the Faculty Staff.

4. **Deployment plan**
    a. Define deployment procedures and timelines based on the agreed project schedule and requirements.
    b. Coordinate with the Dean's Office Staff to schedule deployment of the web-based record management and tracking system and minimize disruption to users.
    c. Prepare deployment scripts and automation tools to streamline the deployment process to the live server.

5. **User Training support**
    a. Create training materials and conduct training sessions for the Administrator, Department Chairpersons, and Faculty Staff of CEIT.
    b. Provide ongoing support and assistance to users during the transition period.
    c. Establish communication for users to report issues and request assistance.

6. **Post deployment evaluation**
    a. Conduct post-deployment reviews to assess the success of the implementation and identify areas for improvement.
    b. Gather feedback from users to evaluate user satisfaction and identify some issues and concerns.
    c. Deal with any post-deployment issues or bugs through patches, updates, or hotfixes as necessary.

7. **Documentation and knowledge transfer**
    a. Update the documentation and knowledge repositories with any changes or updates that happen during the implementation process of the system.
    b. Conduct knowledge transfer sessions to ensure that the CEIT Staff are familiar with the implemented new system.
    c. Document every lesson learned and technical best practices and standard for future reference and the continuous improvement of the system.

8. **Closure and Handover**
    a. Obtain sign-off from the Dean's Office to formally close the implementation phase.
    b. Hand over responsibility for ongoing maintenance, support, and operations to the appropriate teams or personnel.
    c. Conduct a post-implementation review to document key learnings and identify areas for improvement in future projects.

To fully support this implementation plan, specific overall resources will be required.

a. Hardware: The implementation will utilize the existing desktop computers and laptops already present within the College of Engineering and Information Technology (CEIT). No additional hardware procurement is required, as the system will be hosted on the

institution's existing internal server infrastructure managed by the CEIT.

b. Software: The implementation requires a modern web browser (e.g., Google Chrome, Mozilla Firefox) for client-side access, alongside a server-side environment running PHP and MySQL to support the CollTrace system. All required software dependencies and libraries will be installed and configured to align with the existing versions currently supported by the CEIT infrastructure.

c. Facilities and Personnel: The deployment will take place physically at the College of Engineering and Information Technology, Cavite State University - Main Campus. It will involve the researchers, the designated System Administrator, the Department Chairpersons, and the Faculty Staff of CEIT who will act as the primary evaluators and end users of the system.

d. Site-Specific Requirements: A stable internet connection within the CEIT facility is required to access the system during training and evaluation. Since internet connectivity is already provided by the institution, no additional cost for internet services will be incurred.

As detailed in the preceding Budgetary Requirements section, the successful deployment and implementation of the proposed CollTrace system is estimated to cost ₱6,600. This budget will actively support the implementation phase by funding the printing of the final user manuals for the

Administrator, Department Chairpersons, and Faculty Staff, as well as the hardbound documentation required for the formal closure and handover. Since the system will be hosted on CEIT's existing internal server infrastructure, internet connectivity is already provided by the institution, and all training and coordination activities will be conducted on-campus, no additional expenses for web hosting, domain registration, or internet services are required, allowing the allocated budget to be directed solely toward the documentation and printing needs essential to a successful implementation.

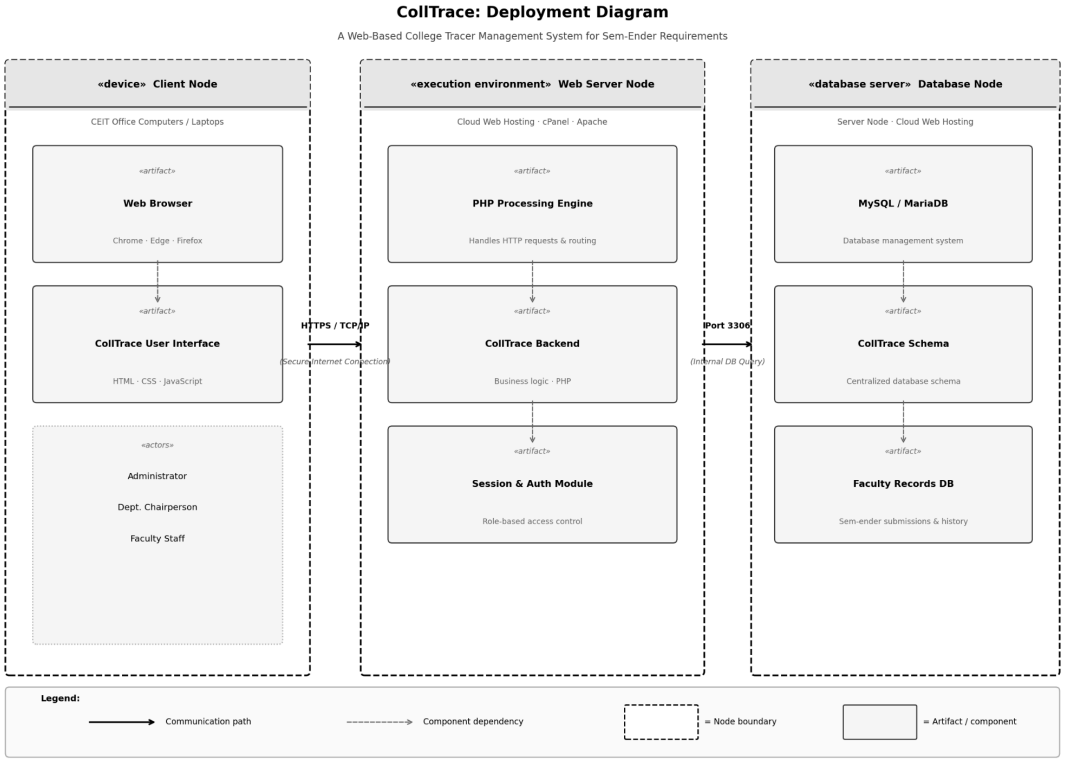



Figure 28. Deployment diagram

The physical and logical hardware configuration required to run the CollTrace: A Web-Based College Tracer Management System in a live environment. The architecture follows a standard 3-tier client-server model, utilizing the resources outlined in the project’s budgetary requirements.

The architecture consists of two primary hardware nodes:

**1. Client Node (CEIT Office Computers / Laptops):** This node represents the existing hardware resources located at the Cavite State University College of Engineering and Information Technology office. Faculty Staff, Department Chairpersons, and Administrators do not need to install specialized software; they will access the system's graphical interface utilizing standard web browsers (e.g., Google Chrome, Microsoft Edge, Mozilla Firefox).

**2. Server Node (Cloud Web Hosting):** Representing the ₱3,000.00 annual web hosting allocation, this node acts as the centralized repository of the system. It is subdivided into two execution environments:

**a. Web Server:** This environment executes the core PHP backend logic, including the role-based access control and submission tracking modules, and securely serves the web pages to the clients.

**b. Database Server:** This environment runs the MySQL/MariaDB database engine, physically storing the centralized database schema, which includes all user credentials, faculty submission records, folder directories, and compliance report data.

Data transmission between the Client Node and the Server Node is strictly routed over the internet using Hypertext Transfer Protocol Secured (HTTPS). This ensures that sensitive university data and login credentials are encrypted in transit. Internally, the Web Server communicates with the Database Server utilizing standard port 3306 for rapid, secure SQL query execution.

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# APPENDICES

## APPENDIX TABLES

Appendix Table 1. Profile of Respondents in Dean's Office

| Dean's Office | |
|---|---|
| **Name of Respondent** | **Position** |
| Sir. Bryan Allen Perido | |

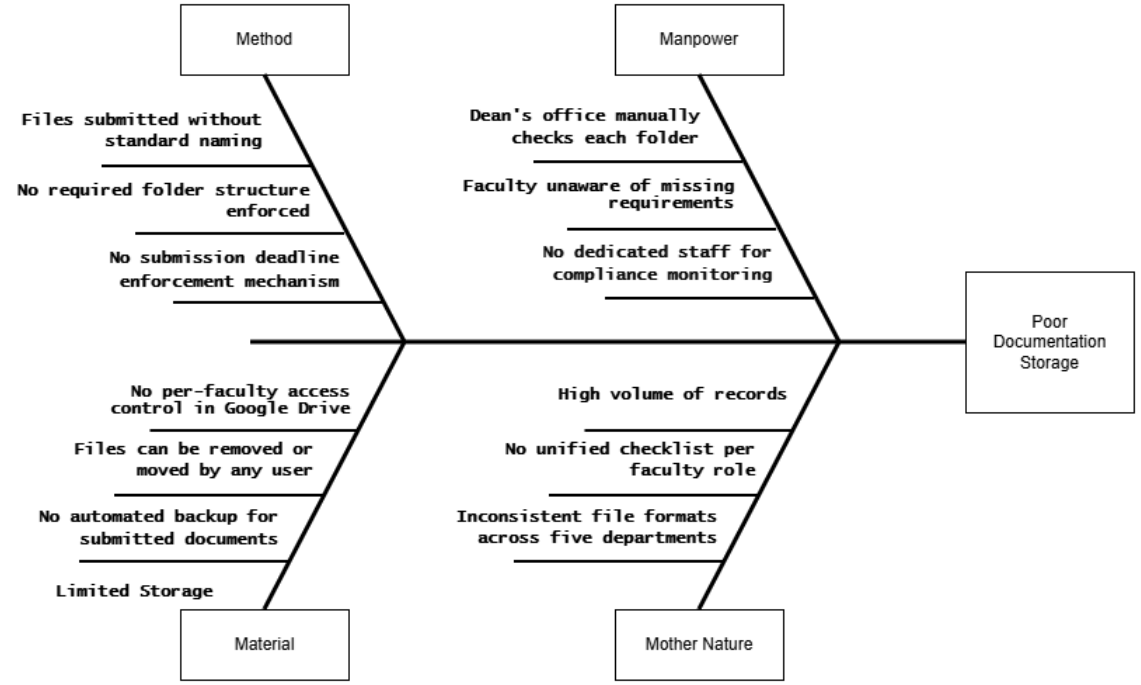
Appendix Table 2. Profile of Respondents of Faculty Chairpersons

| Department Chairperson | |
|---|---|
| **Name of Respondent** | **Position** |
| Ma'am Charlotte B. Carandang | Chairperson, DIT |
| Sir. Danielito R. Escaño | Chairperson, DIET |
| Sir. Michael T. Costa | Chairperson, DCEEE |
| | Chairperson, DAFE |

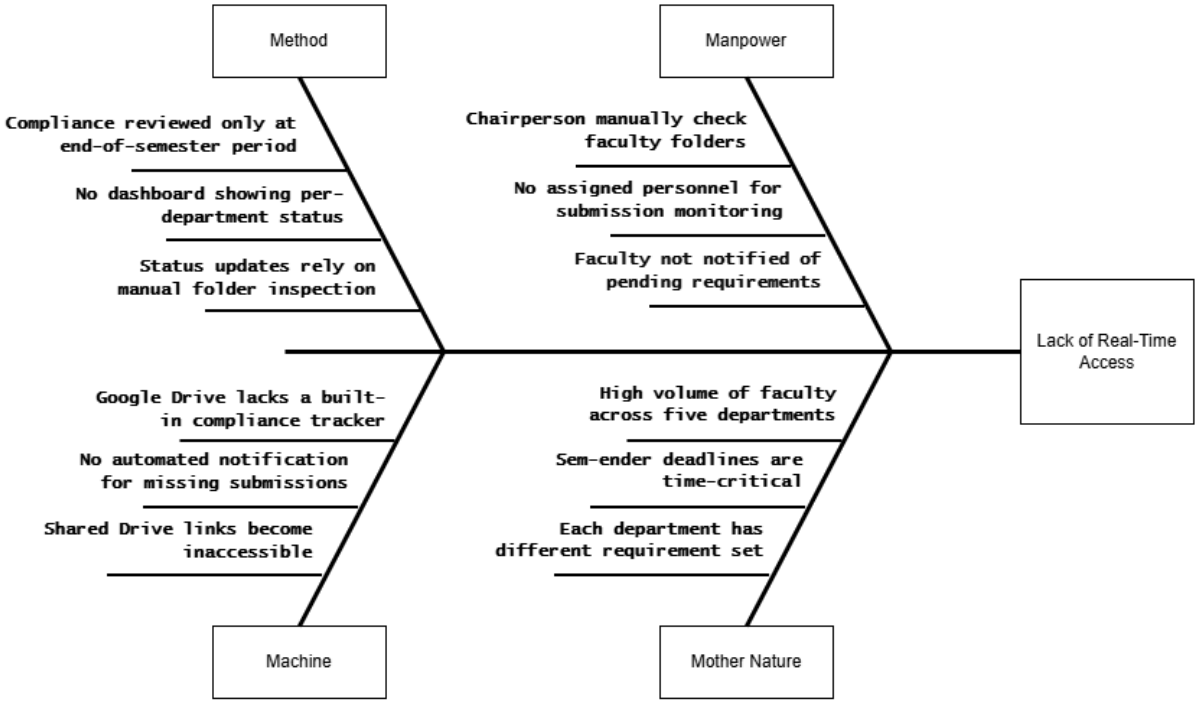
Appendix Table 3. Profile of Respondents from some of Departments in CEIT

| Faculty Staff | |
|---|---|
| **Name of Respondent** | **Department** |
| Respondent 1 | DIET |
| Respondent 2 | DAFE |
| Respondent 3 | DAFE |

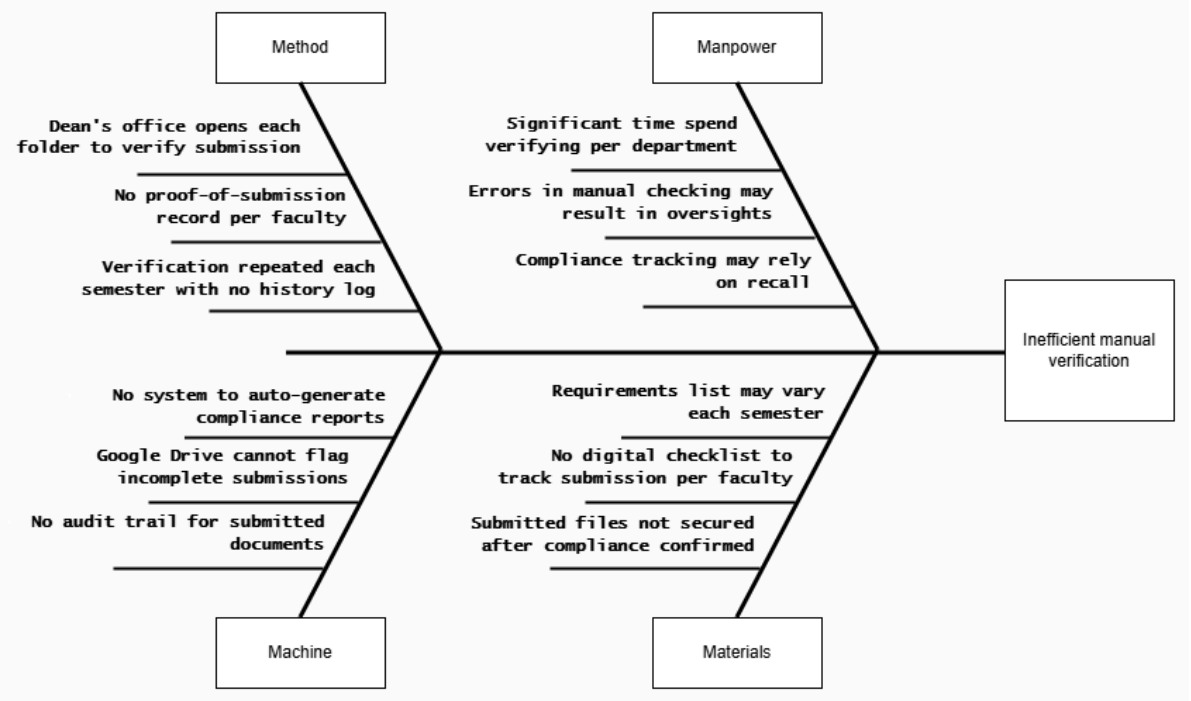
## APPENDIX FIGURES




Appendix Figure 1. Fishbone Diagram for Poor Documentation Storage

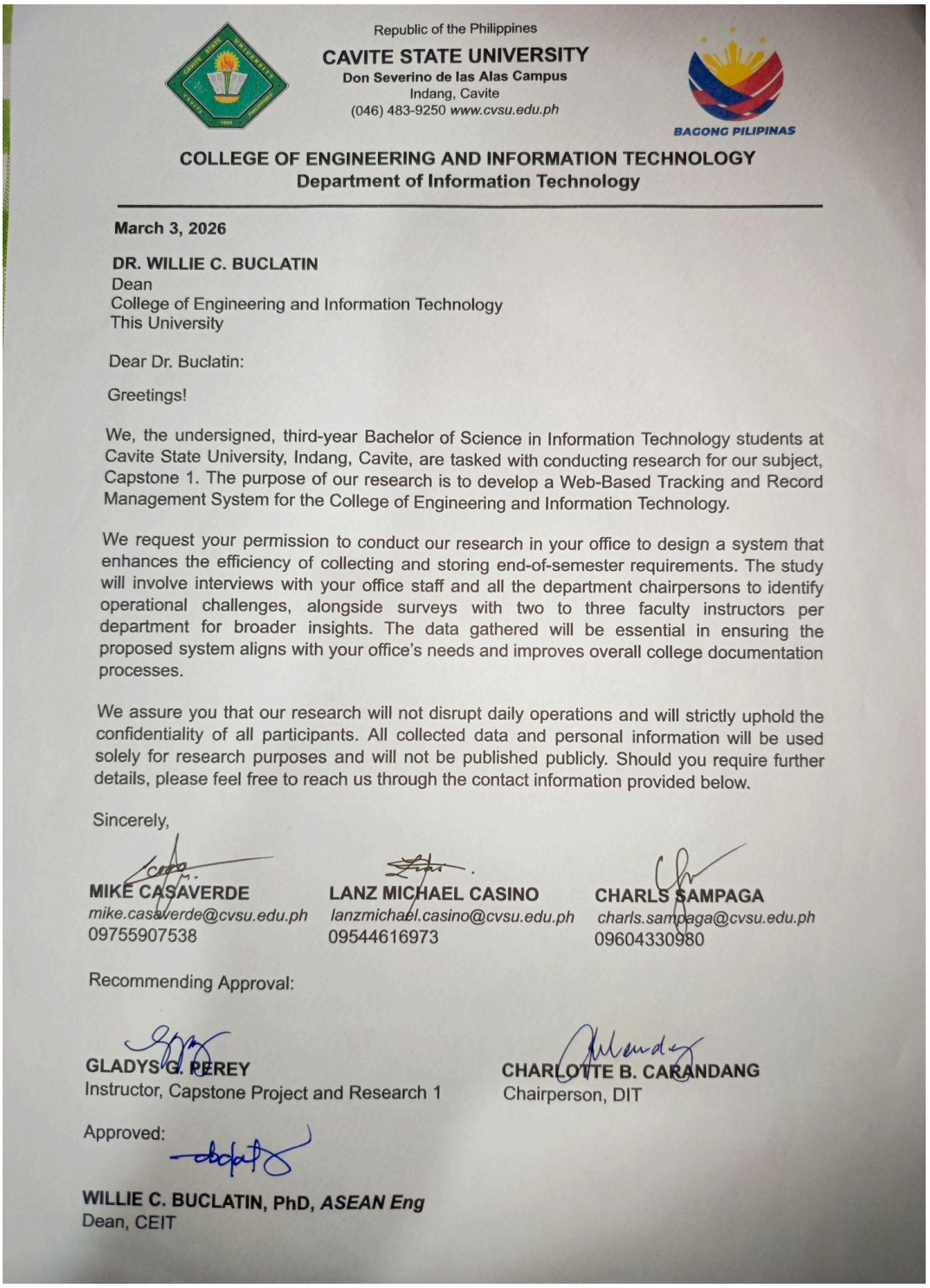



Appendix Figure 2. Fishbone Diagram for Lack of Real Time Tracking




Appendix Figure 3. Fishbone Diagram for Insufficient Manual Vaerification

## OTHER APPENDICES



Republic of the Philippines
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BAGONG PILIPINAS

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

**March 3, 2026**

**DR. WILLIE C. BUCLATIN**
Dean
College of Engineering and Information Technology
This University

Dear Dr. Buclatin:

Greetings!

We, the undersigned, third-year Bachelor of Science in Information Technology students at Cavite State University, Indang, Cavite, are tasked with conducting research for our subject, Capstone 1. The purpose of our research is to develop a Web-Based Tracking and Record Management System for the College of Engineering and Information Technology.

We request your permission to conduct our research in your office to design a system that enhances the efficiency of collecting and storing end-of-semester requirements. The study will involve interviews with your office staff and all the department chairpersons to identify operational challenges, alongside surveys with two to three faculty instructors per department for broader insights. The data gathered will be essential in ensuring the proposed system aligns with your office's needs and improves overall college documentation processes.

We assure you that our research will not disrupt daily operations and will strictly uphold the confidentiality of all participants. All collected data and personal information will be used solely for research purposes and will not be published publicly. Should you require further details, please feel free to reach us through the contact information provided below.

Sincerely,

**MIKE CASAVERDE**
*mike.casaverde@cvsu.edu.ph*
09755907538

**LANZ MICHAEL CASINO**
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Recommending Approval:

**GLADYS G. PEREY**
Instructor, Capstone Project and Research 1

**CHARLOTTE B. CARANDANG**
Chairperson, DIT

Approved:

**WILLIE C. BUCLATIN, PhD, *ASEAN Eng***
Dean, CEIT

Appendix 1. Approved letter for data gathering

DCEEE

Interviewee (Tagalog): Ito, masasagot ito ng ano, pare-parehas lang kasi ito per department. At iisa lang ang process nito, which is ipapasa lang online.

Interviewer (Tagalog): Sa Google Drive po?

Interviewee (Tagalog): Oo, sa Google Drive lang lahat 'yan. Ang ginagawa lang namin is chine-check lang namin kung kumpleto na 'yung files, parang gano'n.

Interviewee (Tagalog): Ah, sila ang naglalagay nito. Ang naglalagay nito ay 'yung sa Dean's Office. Nakalagay na, pagka-naka-tap ng mga faculty, nandoon na 'yung mga pangalan nila, nandoon na per department, tapos 'yung pangalan nandoon na.

Interviewer (English): How often do you need this system to accommodate other reports that are not part of the standard?

Interviewee (Tagalog): Wala, kung ano lang ang nandoon, 'yun lang ang nilalagay namin. Hindi na kami nagdadagdag pa kasi sobrang dami na noon eh.

Interviewer (English): What are the faculty's most common reasons for late submission?

Interviewee (Tagalog): Kasi hindi pa tapos ang ano—ano, nagche-check pa lang, gano'n. Usually naman gano'n ang nangyayari. Mahaba, masyadong maraming requirements na ginagawa, hindi agad natatapos lahat, chine-checkan.

Interviewee (Tagalog): Ano na 'to, from ano ito, pagka nakagawa na sila ng drive, ipo-post namin sa GC ng Chairperson tapos ifo-forward namin sa GC ng department. Gano'n lang. Basta kung nag-send na sila, isesend na rin namin.

Interviewer (English): How would a dashboard showing the presentation...

Interviewee (Tagalog): Wala, wala kaming ganyan. Isa-isa mo chicheck-in. 'Yun ang gagawin ninyong enhancement.

Interviewer (English): Would an automated notification system for the...

Interviewee (Tagalog): 'Yan, sige, maganda 'yan. Idagdag niyo 'yan.

Interviewer (English): For the last, is there any features that you want to...

Interviewee (Tagalog): 'Yun, ay 'yun nga. Maganda doon ay may monitoring na makikita ay kung ano pa 'yung—kung sino pa ang hindi nakakapasa, aling documents ang kulang pa. Ayun, para 'yung automatic na may notification. Tapos pagka may nadagdag, merong notification na gano'n. 'Yun lang naman.

Interviewee (Tagalog): Oo, 'yun lang. Kasi sa amin, ang problema kasi namin, hindi namin alam kung kailan nag-a-upload. So pagka meron kaming isyung i-checheck na ano, pabalik-balik kami gano'n. Hindi namin alam kung kailan nag-a-upload, walang updating. Kaya mas magandang parang may notification, halimbawa para sa email o kaya parang sa FB na nalamang nag-upload ng file o gano'n. Nag-update ng story gano'n, instead na nag-ano—sa activities nandoon kung ano 'yung nadagdag na file at kung ano pa 'yung kulang. At the same time, gano'n din sa faculty na, "Uy, kulang ka pa," makikita niya, may notification na kulang pa siya. 'Yun 'yung idagdag niyo.

# Appendix 2. Transcript of DCEEE Chairperson's Interview

DAFE

Interviewee (Tagalog): Sige, mag-record na.

Interviewer (English): Okay, so for the first question po: As a primary person responsible for tracking submissions, what is your current manual process for verifying all faculty members in your department have submitted their required documents po?

Interviewee (Tagalog/English): Through G-Drive. Meron kaming ano, 'yung DAFE email, tapos meron siyang Drive. So doon lang chine-check kung complete na ang submissions.

Interviewer (English): So bale po, currently po, how do you manage different submission deadlines or requirements for regular faculty members or COS (Contract of Service) faculty members?

Interviewee (Tagalog/English): Ano, doon pa rin, manually na chine-check through G-Drive. Wala pa kaming something na—'yun nga ang maganda sana eh, na meron siyang tracking.

Interviewee (Tagalog/English): Tapos sana mag-prompt siya kay faculty na kulang pa siya ng ganito. Tapos 'pag na-submit na niya, o kaya kunwari nagre-red na siya, dalawa ang pwedeng i-prompt: si faculty at saka si chair para sa follow-up. Gano'n.

Interviewer (English): Bale po in administrative tasks po na ginagawa niyo po, how much time is currently spent po, for example, adding faculty members po or setting semester tracking lists po?

Interviewee (Tagalog/English): Adding? Ano 'yun? Like 'pag nadadagdagan po ng faculty members or 'pag magdadagdag ng list of requirements?

Interviewee (Tagalog/English): List of requirements? Kasi basically ang ano faculty kasi namin ngayon, ano, kami-kami lang din, wala pang bago. Pero siguro hindi naman siya gano'n ka-time consuming 'pag may bagong faculty, in terms of orientation.

Interviewer (English): The next po is, how often do you need the system to accommodate other reports that are not part of the standard or predefined list po?

Interviewee (Tagalog/English): Ulit?

Interviewer (English): How often do you need the system to accommodate other reports that are not part of the standard or predefined list po?

Interviewee (Tagalog/English): Ah, siguro ano—'yan kasi ang gusto kong mangyari sa department eh, may system. Always. Kasi mas madali siya 'pag may system, doon lang ako titingin tapos hindi na ako mag-manual check, 'di ba? Magpo-prompt na siya sa mga concerned na person, kunwari sa akin at saka sa faculty.

Interviewer (English): So in terms of experience po, what are the faculty's most common reasons for late submissions or not yet submission status within your department po, and how do you currently follow up on this?

Interviewee (Tagalog/English): Ang problem talaga is 'yung workload. Siguro madami kasing ginagawa. Mostly, example, 'yung regular faculty members, apart doon sa mga submissions, meron din kasing iba't ibang assignments. Kunwari sa instruction, sa research, sa extension, parang ang daming ginagawa kaya minsan nakakalimutan siya.

Interviewee (Tagalog/English): Sa pagdating naman sa Contract of Service, ang dami din nilang load pagdating sa teaching kaya minsan medyo na-dedelay din. Pero sa ngayon naman, okay-okay naman ang submissions nila, on time naman.

Interviewer (Tagalog): How do you disseminate information sa department?

Interviewee (Tagalog/English): Sa department? Ano, may GC kami. Tapos 'pag super important na 'yung gusto ko na mabalikan nila, email.

Interviewer (English): How would a dashboard showing the percentage of reports submitted by your department help you during the end-of-semester rush?

Interviewee (Tagalog/English): Very ano, very helpful kung merong gano'n na dashboard. Kasi titingnan ko lang 'yung report at malalaman ko kung sino ang hindi nakapag-submit pa. Very helpful 'yan kung magkakaroon.

Interviewer (English): Aside from monitoring and tracking po, ano sa tingin niyo po 'yung pinaka-other—bukod po doon, ano pa po 'yung gusto niyo po maging feature sa system?

Interviewee (Tagalog/English): Repository. 'Yung lahat ng form na kailangan namin, nandoon na. Tapos lahat ng syllabus nandoon na din na pwede na lang i-ano ni, kunwari bagong hire si faculty, hindi na siya magtatanong doon sa kunwari umalis na faculty or old faculty. Nandoon na lahat sa dashboard kasi kasama 'yun sa end-of-sem na requirement.

Interviewer (English): Next naman po is, would an automated notification system for deadlines reduce the need for manual follow-ups, and what frequency should these alerts be sent?

Interviewee (Tagalog/English): Mas maganda kasi very helpful 'yung notification. Email. Kasi 'pag SMS kasi, medyo ano na siya, out of boundary na sa personal ni faculty 'yun.

Interviewer (Tagalog/English): Aside from repository, meron pa po ba kayong edge ng system na ina-expect niyo na gawin nung system for you to be able to be effective in terms of administrating the ano?

Interviewee (Tagalog/English): Ma'm Mel, example meron tayong dashboard, gagawan nila tayo. Ano bang magandang feature aside sa mga repository natin, aside sa reports? 'Yung sa progress ng bawat ano, activities 'no? 'Yung parang ano, style na SmartSheet, alam niyo 'yun? Gano'n. Meron kaming plano na activity tapos nandoon 'yung mga tao na involved, responsible person. Kunwari sa paggawa ng proposal, mga ganyan. Tapos i-ti-tick namin, kunwari mag-set kami ng start date at saka end date.

Interviewee (Tagalog/English): Tapos after mag-end 'yung isang responsible person, magpo-prompt kay next na, "Ikaw na, napasa na niya ang bola." Magpo-prompt ng email sa kanya na ikaw na ang kasunod. Parang progress tracker siya, kung familiar kayo sa SmartSheet.

# Appendix 3. Transcript of DAFE Chairperson's Interview

DIET

Interviewer (English): This interview is for research purposes. So number one question po under po sa current situation niyo po ng pag-submit: As a primary person responsible for tracking submissions, what is your current manual process for verifying that all faculty members in your department have submitted their required documents?

Interviewee (English/Tagalog): Manual process? Current manual process? Actually, merong Drive na nilalagyan. And then, before I sign their clearance every end of the sem, regardless of the—ano bang tawag namin—kung siya ay JO (Job Order) or permanent na faculty member, talagang religiously nagpapasa sila. Then 'yan, kung makikita niyo parang basket ko diyan, 'yung laman ng Drive ini-screen ko para I have the proper documentation na 'yun 'yung file na naka-upload doon sa Drive.

Interviewer (Tagalog): Meron pa rin po kayong parang nagpi-print po kayo for this?

Interviewee (English/Tagalog): Parang additional evidence lang siya bukod doon sa Drive para may hawak akong hard copy, para evident talaga. At doon pala, na-track at na-trace.

Interviewer (English/Tagalog): So number two is: How do you currently manage the different submissions deadlines or requirements for regular compared to COS (Contract of Service) faculty members?

Interviewee (English/Tagalog): To tell you honestly, mas mabilis mag-submit ang COS or JO.

Interviewer (English/Tagalog): Ayun nga po, 'di ba ang COS po is every sem po sila may requirement po talaga?

Interviewee (English/Tagalog): We all do our part religiously. Nagpapasa naman. There are just only some delays

Interviewer (English): Next question po is: How much time is currently spent on administrative tasks, for example, adding new faculty members or setting up a new semester's tracking list?

Interviewee (English/Tagalog): Actually me ha, I hate beating the deadline. I make sure na ahead of time, prepared na ako. Kasi ako, if I may share, I've been the director doon sa Physical Plant Services for seven years, and my managerial skills—isipin mo hawak ko is 69 non-academic staff, maintenance ng buong university. That's why 'yun ang natutunan ko, that I have to be systematic. Kasi hindi po pwede 'yung, "Ito na, wala pa ako nito". I make sure na 'yung resources ay nandoon before dumating sa deadline.

Interviewer (English): So 'yung next po is: How often do you need the system to accommodate other reports that are not part of the standard or predefined list?

Interviewee (English/Tagalog): Marami kasing importance when it comes to accreditation. 'Yung lahat ng nire-require din sa faculty, each part of the documents needed during level accreditation ng program, talagang napakalaking bagay. Kaya itong naisip ng kolehiyo na end-of-sem requirements, it's a big ease during accreditations. Why? Hindi ka na maghahanap 'pag malapit na ang accreditation, andoon na sa Drive. Mas maganda actually, it's standardized and centralized.

Interviewer (English): In terms of experience po: What are the faculty's most common reasons for late submissions or "yet to submit" status within your department, and how do you currently follow up on this?

Interviewee (English/Tagalog): The common reason for me ha, siguro 'yung regular faculty members ay hindi lang naman ito ang ginagawa. We have a lot of lateral assignments. Hindi gaya ng COS, they just teach and go. Unlike kami, marami talaga. Bukod sa accreditation, marami pa kaming committees sa university-wide na kami ay kasali. Kaya you cannot blame also the regular faculty members na hindi makasubmit on time. And that's nasa adjustment na ng Dept. Chair how to follow up.

Interviewer (Tagalog): How do you disseminate information sa department?

Interviewee (English/Tagalog): Informations were disseminated doon sa GC (Group Chat). And also merong office memo na ini-issue para ang standard pa rin pagdating sa supervisory level na ganyan ay may memo na signed, na they have read and seen the memo.

Interviewer (English/Tagalog): Helpful po ba na maglagay kami ng parang announcement tab, like 'yung mga memos or department chair memos?

Interviewee (Tagalog): Oo naman, would be helpful din po.

Interviewer (English): How would a dashboard showing the percentage of reports submitted by your department help you during the end-of-semester rush?

Interviewee (English/Tagalog): Actually, 'yung dashboard is just an additional, pero ako personally, pagtingin ko pa lang doon sa Drive, kasi naka-separate naman, kaya na. Pero kung additional 'yung dashboard, mas okay.

Interviewer (English): Would an automated notification system for deadlines reduce the need for manual follow-ups, and at what frequency should these alerts be sent?

Interviewee (Tagalog): Helpful talaga siya. 'Yung frequency, kayo na mag-decide.

Interviewer (English): Last po: Is there any features that you want to add or expect in the proposed system na would be beneficial for you?

Interviewee (English/Tagalog): Ako honestly, sabi ko nga, doon sa experience ko for seven years, marami na akong natulungan na IT/CS students when it comes sa operations ng PPS before. Sabi ko nga, nung nailatag sa akin 'yung finished product, that's the only time na nagkokomentaryo ako: dagdagan mo rito, ganyan. And I'm helping the students kung paano i-improve.

Interviewer (English/Tagalog): Additionally po, since some faculty members handle the same subjects, would it be helpful if they only upload one syllabus instead of duplicates for each staff?

Interviewee (English/Tagalog): Actually kami, let's say same 'yung subject, nag-a-upload din ako ng mga requirements, hindi iisa lang. Kunwari common kami sa syllabus, nag-a-upload din ako ng sa akin separately. Kasi nakikita doon sa folder kung 'yung activities mo at TOS (Table of Specifications) ng exam mo ay tama doon sa time table ng syllabus mo. 'Yun 'yung isang hinuhuli ng accreditor. It's a big explanation 'pag nandoon na ang accreditor. Pero dapat 'yung 18 weeks, ma-comply mo lahat ng topics and deliverables.

Interviewer (Tagalog): Ayun lang po. Maraming salamat po.

Interviewee (Tagalog): Sige po, thank you.

# Appendix 4. Transcript of DIET Chairperson's Interview

DIT

Interviewer: So, we're just here to ask about the end-of-semester requirements that are submitted by every department.

Chairperson: Okay, so what are your questions?

Interviewer: For number one, it's about the current situation of submissions. As the primary person responsible for tracking submissions, what is your current manual process for verifying that all faculty members in your department have submitted their required documents?

Chairperson: Okay, so actually, there's a followed process or procedure in the college, which is localized only to the CIT. So, before the end of the semester, they provide a Google Drive link. There's a folder for each academic year and per semester, and that's where all the end-semester requirements are placed, both for regular and contract of service (COS) faculty. So, there is no manual procedure. But the process would be a drive link where the faculty members upload individually—they have their own folders. And then, the CEIT also has an internal drive. So, it's up to the department right now if we want to have a duplicate copy. But for the last two to three semesters, it hasn't been as usable, or we haven't practiced it. Because it's internal, it's our option if the department will continue using that.

Interviewer: So, the next one is: how do you currently manage the different submission deadlines or requirements for regular versus COS faculty members?

Chairperson: There's actually no difference. If there is one, it's that COS faculty have their clearance every semester. For us in the regular faculty, for whoever is on teacher's leave—which happens after the second semester—that's the only time they process their clearance. So, that's the only difference. But in terms of requirements, regardless of tenure, they are required to submit the end-semester requirements.

Interviewer: So, there's no significant difference?

Chairperson: No significant difference. You might just notice it in the timeliness of the submission. It is already given that regular faculty members have other assignments aside from teaching. So, there are chances or possibilities that they cannot submit the requirements [as quickly] compared to the COS, whose job or responsibility is mainly to teach. In terms of timeliness, we can expect the COS [to be faster] because of the clearance requirement; they really need to submit the requirements because they cannot claim their last salary without the clearance. So, I think that's the biggest motivation. But with respect to our regular faculty members, even though their submission isn't as early, we have to situate that all pertinent records will be available in case there is a need for them for other accreditation or whatever visits—so everything is available.

Interviewer: Okay. Next is: how much time is currently spent on administrative tasks such as adding new faculty members or setting up a new semester's tracking?

Chairperson: I think the question should be: what is the system being used for that first? As I mentioned, we are just using Google Drive. So, there's no such thing as registration or anything like that. Once you are a faculty member—as I said, they already provided a direct link—therefore, you already have a folder under your name. So, we don't have a process for adding faculty. For example, me as faculty, I don't add myself, nor do I add at the department level. It is a responsibility of the staff or clerk of the Dean's Office to ensure that every faculty member here in the CEIT will have a separate Google Drive folder where they will put their end-semester requirements.

Chairperson: Yes, exactly. And perhaps an automated reminder that goes directly to the faculty who haven't finished yet. Instead of me posting manually in the group chat, the system could just send them a nudge.

Interviewer: That makes a lot of sense. Do you have any other suggestions or features you'd like to see in a new system?

Chairperson: Just ease of use. Most of our faculty members are already comfortable with the current setup, so any new system shouldn't be too complicated. It should be user-friendly, especially for those who aren't as tech-savvy. If it's a mobile-friendly app or website, that would be even better so they can check their status on the go.

Interviewer: Noted on that. How do you think an automated system would change the culture of submission in the department?

Chairperson: I think it would promote more accountability. If everyone can see—not necessarily everyone, but if there's a clear record that is easily accessible—faculty members might be more inclined to submit on time. It removes the "I forgot" excuse because the system is there to remind them. It also gives me more time to focus on other departmental concerns rather than just being a "reminder machine."

Interviewer: (Laughs) A reminder machine. That's a good way to put it. Well, that's all for our questions. We really appreciate you taking the time to talk with us today.

Chairperson: No problem. I'm looking forward to seeing what you come up with. If you need anything else, just let me know.

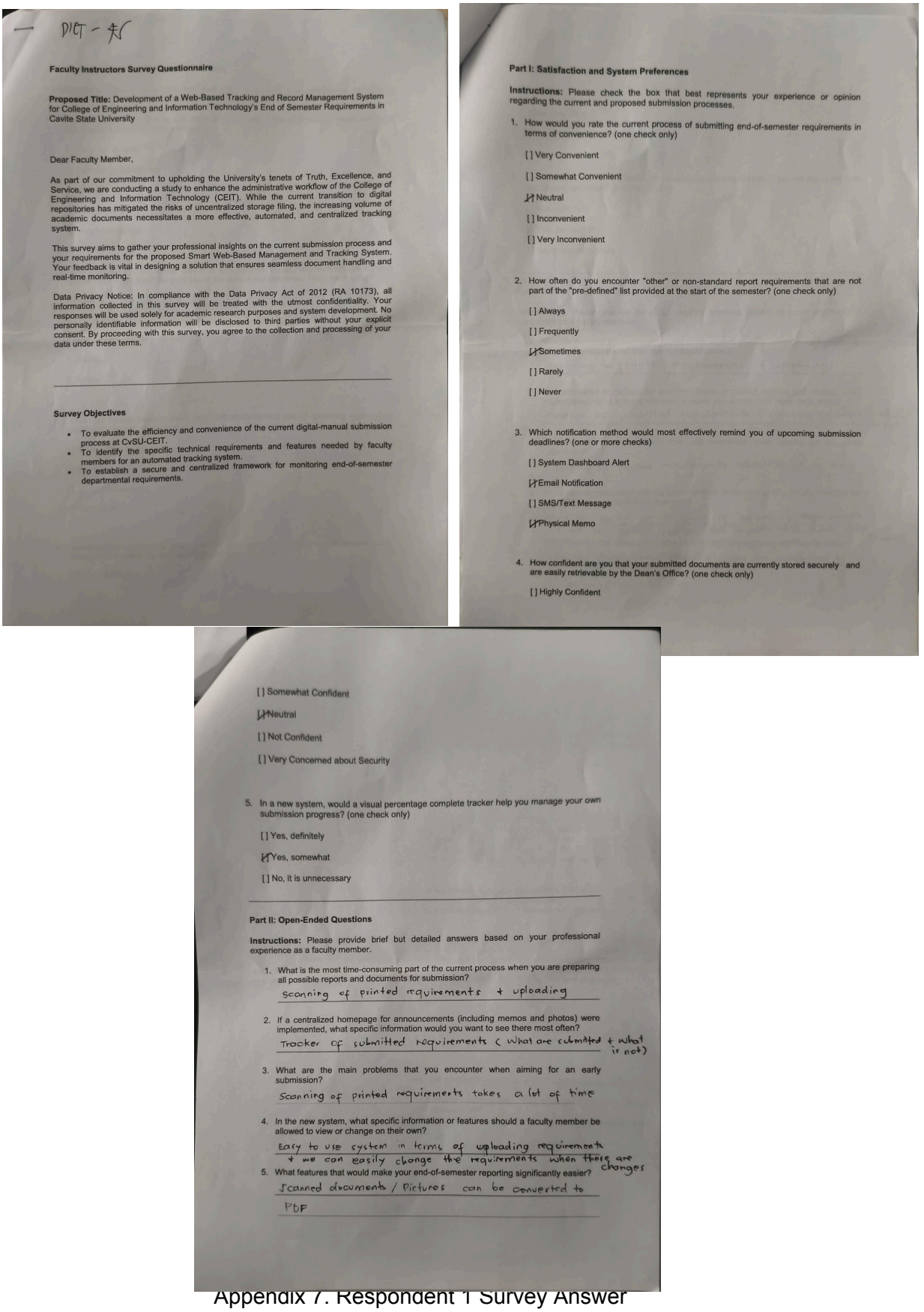
Interviewer: We will. Thank you!

Chairperson: You're welcome.

Appendix 5. Transcript of DIT Chairperson's Interview




Appendix 6. Proof of Interview



DIET - BS

**Faculty Instructors Survey Questionnaire**

**Proposed Title:** Development of a Web-Based Tracking and Record Management System for College of Engineering and Information Technology's End of Semester Requirements in Cavite State University

Dear Faculty Member,

As part of our commitment to upholding the University's tenets of Truth, Excellence, and Service, we are conducting a study to enhance the administrative workflow of the College of Engineering and Information Technology (CEIT). While the current transition to digital repositories has mitigated the risks of uncentralized storage filing, the increasing volume of academic documents necessitates a more effective, automated, and centralized tracking system.

This survey aims to gather your professional insights on the current submission process and your requirements for the proposed Smart Web-Based Management and Tracking System. Your feedback is vital in designing a solution that ensures seamless document handling and real-time monitoring.

Data Privacy Notice: In compliance with the Data Privacy Act of 2012 (RA 10173), all information collected in this survey will be treated with the utmost confidentiality. Your responses will be used solely for academic research purposes and system development. No personally identifiable information will be disclosed to third parties without your explicit consent. By proceeding with this survey, you agree to the collection and processing of your data under these terms.

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**Survey Objectives**

- To evaluate the efficiency and convenience of the current digital-manual submission process at CvSU-CEIT.
- To identify the specific technical requirements and features needed by faculty members for an automated tracking system.
- To establish a secure and centralized framework for monitoring end-of-semester departmental requirements.

**Part I: Satisfaction and System Preferences**

**Instructions:** Please check the box that best represents your experience or opinion regarding the current and proposed submission processes.

1. How would you rate the current process of submitting end-of-semester requirements in terms of convenience? (one check only)

   [ ] Very Convenient

   [ ] Somewhat Convenient

   [x] Neutral

   [ ] Inconvenient

   [ ] Very Inconvenient

2. How often do you encounter "other" or non-standard report requirements that are not part of the "pre-defined" list provided at the start of the semester? (one check only)

   [ ] Always

   [ ] Frequently

   [x] Sometimes

   [ ] Rarely

   [ ] Never

3. Which notification method would most effectively remind you of upcoming submission deadlines? (one or more checks)

   [ ] System Dashboard Alert

   [x] Email Notification

   [ ] SMS/Text Message

   [x] Physical Memo

4. How confident are you that your submitted documents are currently stored securely and are easily retrievable by the Dean's Office? (one check only)

   [ ] Highly Confident

   [ ] Somewhat Confident

   [x] Neutral

   [ ] Not Confident

   [ ] Very Concerned about Security

5. In a new system, would a visual percentage complete tracker help you manage your own submission progress? (one check only)

   [ ] Yes, definitely

   [x] Yes, somewhat

   [ ] No, it is unnecessary

---

**Part II: Open-Ended Questions**

**Instructions:** Please provide brief but detailed answers based on your professional experience as a faculty member.

1. What is the most time-consuming part of the current process when you are preparing all possible reports and documents for submission?

   Scanning of printed requirements + uploading

2. If a centralized homepage for announcements (including memos and photos) were implemented, what specific information would you want to see there most often?

   Tracker of submitted requirements ( what are submitted + what is not)

3. What are the main problems that you encounter when aiming for an early submission?

   Scanning of printed requirements takes a lot of time

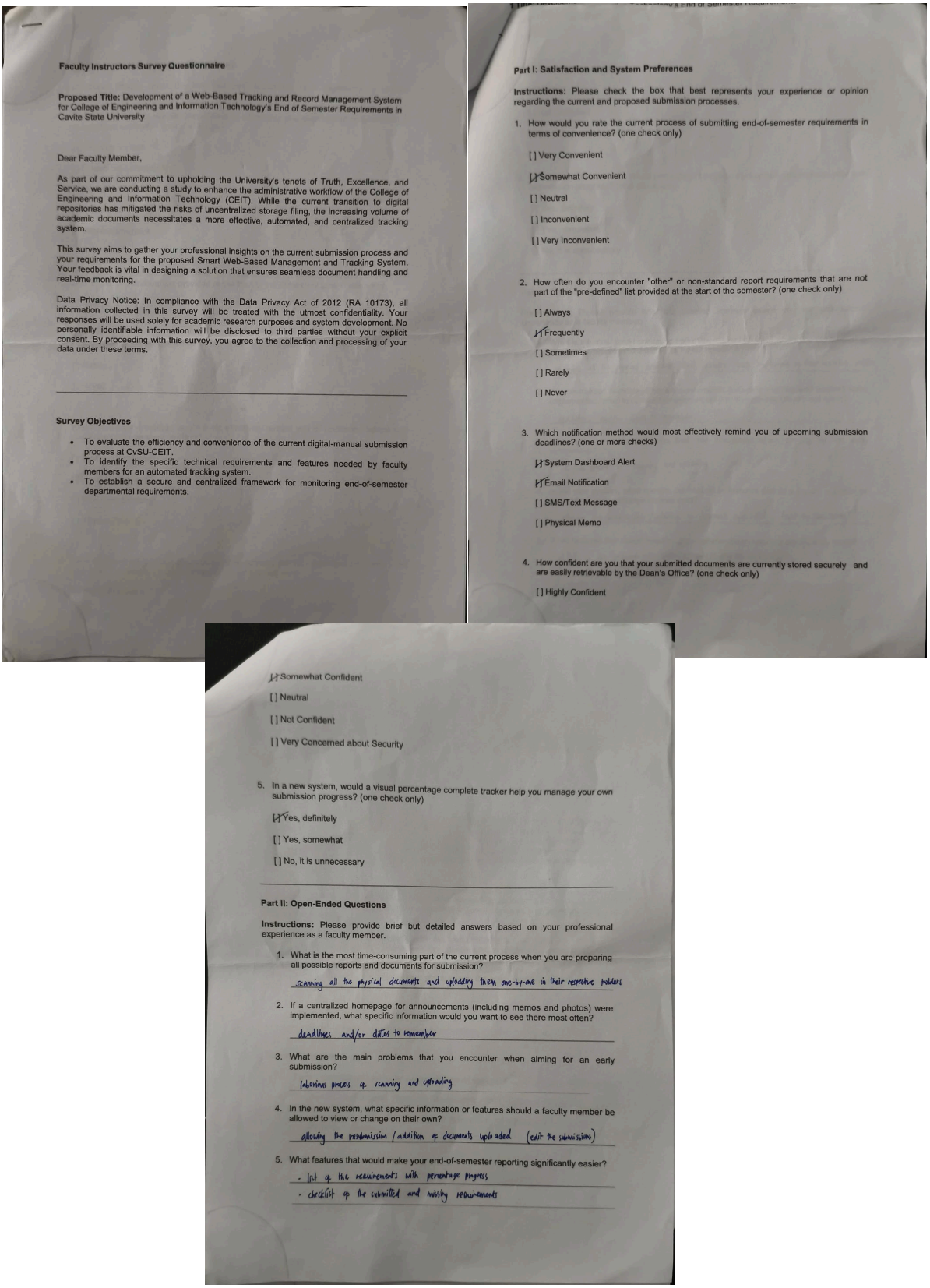
4. In the new system, what specific information or features should a faculty member be allowed to view or change on their own?

   Easy to use system in terms of uploading requirements + we can easily change the requirements when there are changes

5. What features that would make your end-of-semester reporting significantly easier?

   Scanned documents / Pictures can be converted to PDF

Appendix 7. Respondent 1 Survey Answer



**Faculty Instructors Survey Questionnaire**

**Proposed Title:** Development of a Web-Based Tracking and Record Management System for College of Engineering and Information Technology's End of Semester Requirements in Cavite State University

Dear Faculty Member,

As part of our commitment to upholding the University's tenets of Truth, Excellence, and Service, we are conducting a study to enhance the administrative workflow of the College of Engineering and Information Technology (CEIT). While the current transition to digital repositories has mitigated the risks of uncentralized storage filing, the increasing volume of academic documents necessitates a more effective, automated, and centralized tracking system.

This survey aims to gather your professional insights on the current submission process and your requirements for the proposed Smart Web-Based Management and Tracking System. Your feedback is vital in designing a solution that ensures seamless document handling and real-time monitoring.

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---

**Survey Objectives**

- To evaluate the efficiency and convenience of the current digital-manual submission process at CvSU-CEIT.
- To identify the specific technical requirements and features needed by faculty members for an automated tracking system.
- To establish a secure and centralized framework for monitoring end-of-semester departmental requirements.

**Part I: Satisfaction and System Preferences**

**Instructions:** Please check the box that best represents your experience or opinion regarding the current and proposed submission processes.

1. How would you rate the current process of submitting end-of-semester requirements in terms of convenience? (one check only)
   - [ ] Very Convenient
   - [x] Somewhat Convenient
   - [ ] Neutral
   - [ ] Inconvenient
   - [ ] Very Inconvenient
2. How often do you encounter "other" or non-standard report requirements that are not part of the "pre-defined" list provided at the start of the semester? (one check only)
   - [ ] Always
   - [x] Frequently
   - [ ] Sometimes
   - [ ] Rarely
   - [ ] Never
3. Which notification method would most effectively remind you of upcoming submission deadlines? (one or more checks)
   - [x] System Dashboard Alert
   - [x] Email Notification
   - [ ] SMS/Text Message
   - [ ] Physical Memo
4. How confident are you that your submitted documents are currently stored securely and are easily retrievable by the Dean's Office? (one check only)
   - [ ] Highly Confident
   - [x] Somewhat Confident
   - [ ] Neutral
   - [ ] Not Confident
   - [ ] Very Concerned about Security
5. In a new system, would a visual percentage complete tracker help you manage your own submission progress? (one check only)
   - [x] Yes, definitely
   - [ ] Yes, somewhat
   - [ ] No, it is unnecessary

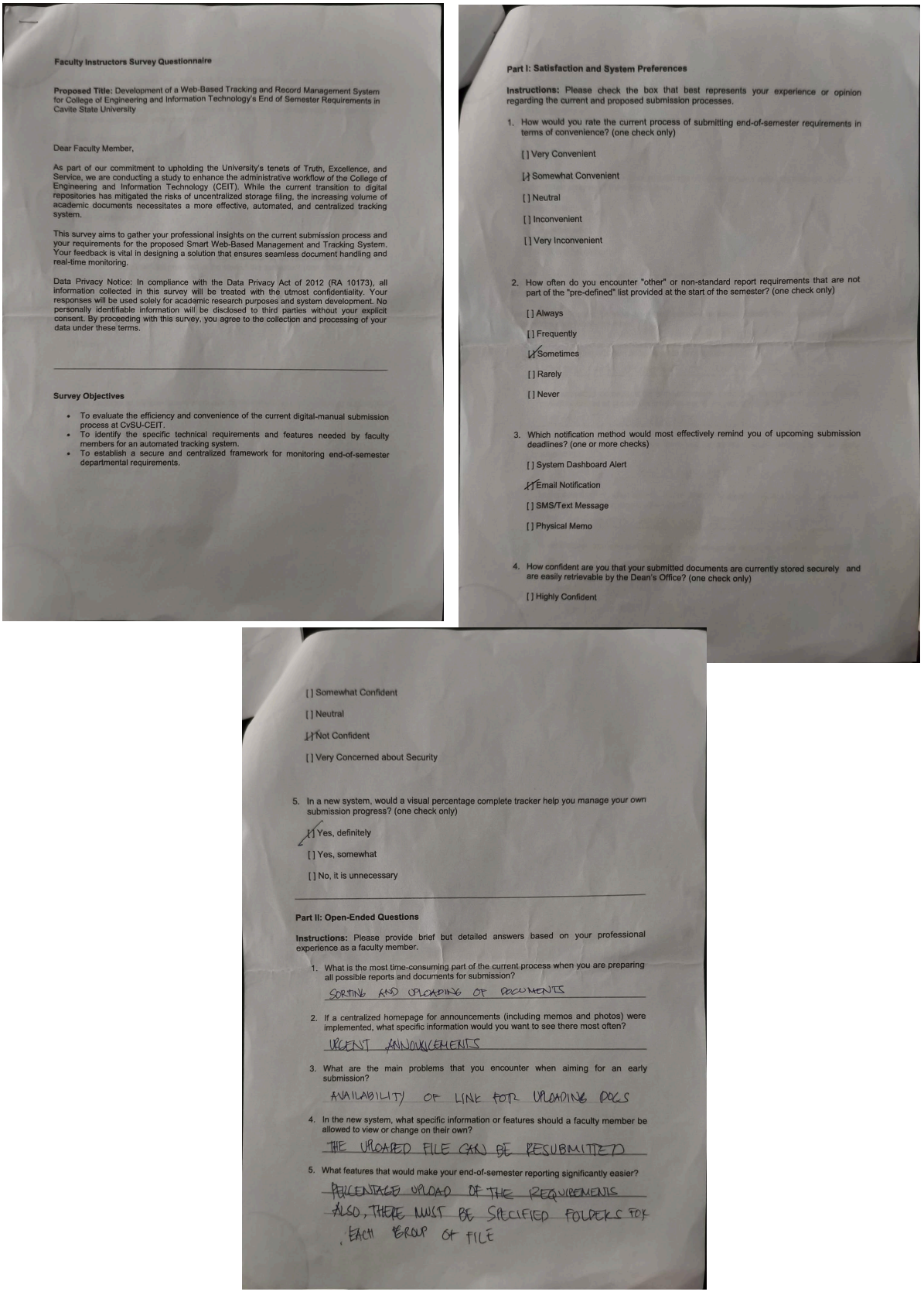
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**Part II: Open-Ended Questions**

**Instructions:** Please provide brief but detailed answers based on your professional experience as a faculty member.

1. What is the most time-consuming part of the current process when you are preparing all possible reports and documents for submission?
   scanning all the physical documents and uploading them one-by-one in their respective folders
2. If a centralized homepage for announcements (including memos and photos) were implemented, what specific information would you want to see there most often?
   deadlines and/or dates to remember
3. What are the main problems that you encounter when aiming for an early submission?
   laborious process of scanning and uploading
4. In the new system, what specific information or features should a faculty member be allowed to view or change on their own?
   allowing the resubmission / addition of documents uploaded (edit the submissions)
5. What features that would make your end-of-semester reporting significantly easier?
   - list of the requirements with percentage progress
   - checklist of the submitted and missing requirements

Appendix 8. Respondent 2 Survey Answer



**Faculty Instructors Survey Questionnaire**

**Proposed Title:** Development of a Web-Based Tracking and Record Management System for College of Engineering and Information Technology's End of Semester Requirements in Cavite State University

Dear Faculty Member,

As part of our commitment to upholding the University's tenets of Truth, Excellence, and Service, we are conducting a study to enhance the administrative workflow of the College of Engineering and Information Technology (CEIT). While the current transition to digital repositories has mitigated the risks of uncentralized storage filing, the increasing volume of academic documents necessitates a more effective, automated, and centralized tracking system.

This survey aims to gather your professional insights on the current submission process and your requirements for the proposed Smart Web-Based Management and Tracking System. Your feedback is vital in designing a solution that ensures seamless document handling and real-time monitoring.

Data Privacy Notice: In compliance with the Data Privacy Act of 2012 (RA 10173), all information collected in this survey will be treated with the utmost confidentiality. Your responses will be used solely for academic research purposes and system development. No personally identifiable information will be disclosed to third parties without your explicit consent. By proceeding with this survey, you agree to the collection and processing of your data under these terms.

---

**Survey Objectives**

- To evaluate the efficiency and convenience of the current digital-manual submission process at CvSU-CEIT.
- To identify the specific technical requirements and features needed by faculty members for an automated tracking system.
- To establish a secure and centralized framework for monitoring end-of-semester departmental requirements.

**Part I: Satisfaction and System Preferences**

**Instructions:** Please check the box that best represents your experience or opinion regarding the current and proposed submission processes.

1. How would you rate the current process of submitting end-of-semester requirements in terms of convenience? (one check only)
   - [ ] Very Convenient
   - [x] Somewhat Convenient
   - [ ] Neutral
   - [ ] Inconvenient
   - [ ] Very Inconvenient
2. How often do you encounter "other" or non-standard report requirements that are not part of the "pre-defined" list provided at the start of the semester? (one check only)
   - [ ] Always
   - [ ] Frequently
   - [x] Sometimes
   - [ ] Rarely
   - [ ] Never
3. Which notification method would most effectively remind you of upcoming submission deadlines? (one or more checks)
   - [ ] System Dashboard Alert
   - [x] Email Notification
   - [ ] SMS/Text Message
   - [ ] Physical Memo
4. How confident are you that your submitted documents are currently stored securely and are easily retrievable by the Dean's Office? (one check only)
   - [ ] Highly Confident
   - [ ] Somewhat Confident
   - [ ] Neutral
   - [x] Not Confident
   - [ ] Very Concerned about Security
5. In a new system, would a visual percentage complete tracker help you manage your own submission progress? (one check only)
   - [x] Yes, definitely
   - [ ] Yes, somewhat
   - [ ] No, it is unnecessary

---

**Part II: Open-Ended Questions**

**Instructions:** Please provide brief but detailed answers based on your professional experience as a faculty member.

1. What is the most time-consuming part of the current process when you are preparing all possible reports and documents for submission?
   SORTING AND UPLOADING OF DOCUMENTS
2. If a centralized homepage for announcements (including memos and photos) were implemented, what specific information would you want to see there most often?
   URGENT ANNOUNCEMENTS
3. What are the main problems that you encounter when aiming for an early submission?
   AVAILABILITY OF LINK FOR UPLOADING DOCS
4. In the new system, what specific information or features should a faculty member be allowed to view or change on their own?
   THE UPLOADED FILE CAN BE RESUBMITTED
5. What features that would make your end-of-semester reporting significantly easier?
   PERCENTAGE UPLOAD OF THE REQUIREMENTS
   ALSO, THERE MUST BE SPECIFIED FOLDERS FOR EACH GROUP OF FILE

Appendix 9. Respondent 3 Survey Answer



*CEIT R&E Form 1*

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
Don Severino delas Alas Campus
Indang, Cavite

***COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY***

**REQUEST FOR ADVISER AND TECHNICAL CRITIC**

NAME: MIKE CASAVERDE, LANZ MICHAEL CASINO, CHARLS SAMPAGA

COURSE: BS INFORMATION TECHNOLOGY MAJOR: _____

AREA OF STUDY: INFORMATION SYSTEMS DEVELOPMENT FOR AN ACTUAL CLIENT

**CONFORME:**

| | | | |
|---|---|---|---|
| **JOY M. PEJI**<br>Adviser | 4.7.26<br>Date | **ROY ALBERT T. LUCERO**<br>Technical Critic | 4/7/26<br>Date |

**RECOMMENDING APPROVAL:**

| | | | |
|---|---|---|---|
| **GLADYS G. PEREY**<br>Unit Research Coordinator | 4-28-26<br>Date | **CHARLOTTE B. CARANDANG**<br>Department Chairperson | 4/28/26<br>Date |

**APPROVED:**

| | |
|---|---|
| **WILLIE C. BUCLATIN**<br>Dean | 28 APR 2026<br>Date |

cc: College R&E Coordinator

Appendix 10. CEIT R&E Form 1